

rainier3d: a reproducible digital model of the subsurface and surface of Mount Rainier, Washington

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Abstract. rainier3d is an open digital model of Mount Rainier and the West Rainier Seismic Zone (WRSZ). It covers a 70×75 km box, from the ground surface to 20 km below sea level. Every input is fetched by a script from its original archive, cached, and recorded with a checksum. Every parameter names its source in a registry, and the whole database can be rebuilt with one environment and a sequence of numbered scripts.

5 The subsurface model holds P- and S-wave speed, density and attenuation on three nested grids. Mapped geology is extended to depth by explicit rules. Near-surface hydrothermal alteration is mapped from a helicopter electromagnetic survey. A pressure-dependent rock-physics law converts the units to seismic properties, and the result is merged with the USGS Cascadia velocity model v1.7 and CRESCENT Gen0 in the wavenumber domain.

10 The model is calibrated on 1823 P and 1280 S analyst picks from 88 Pacific Northwest Seismic Network (PNSN) earthquakes. Each event is relocated in 3D in every trial model, with topography honoured. Three rock-physics multipliers and a depth-dependent correction of the regional models are fitted by Gauss–Newton iterations on P, S and S–P residuals. On held-out events the root-mean-square (RMS) residual falls from 0.121 to 0.093 s for P and from 0.317 to 0.188 s for S. An independent local-earthquake tomography confirms the Vs correction beneath Rainier.

The same grids carry the following surface layers:

- 15
- elevation, geology and glacier thickness;
 - soil thickness, two water-table estimates and streams;
 - land cover, canopy height, leaf area and biomass;
 - satellite imagery.

The model also includes GNSS strain rates and the stress that the weight of the edifice exerts at depth.

20 The derived products are distributed under CC-BY 4.0 with a command-line and Python client. The client writes them in the formats used by eikonal solvers, NonLinLoc, SPECFEM3D and the EarthScope Earth Model Collaboration.

Table 1. Outline of the paper.

Section	Content
Sect. 2	Domain and workflow
Sect. 3	How the data are compiled deterministically, and how to rebuild them
Sect. 4	Surface layers
Sect. 5	Subsurface model, with its calibration and validation
Sect. 6	Strain and stress
Sect. 7	Geohydrology
Sect. 8	Catalogue of mass movements
Sect. 9	Access to the products
Sect. 10	Limitations

1 Introduction

Mount Rainier is the most hazardous volcano in the Cascade Range, for two reasons. Its hydrothermally weakened edifice is covered by glaciers, and its lahars reach populated valleys (Hoblitt et al., 1998; Scott et al., 1995). Seismic monitoring at Rainier relies on one-dimensional velocity models, with station corrections absorbing the effects of topography and near-surface structure. The regional three-dimensional models of Cascadia resolve the crust at wavelengths of several kilometres or more (Wirth et al., 2025; He et al., 2026). They do not resolve the volcanic edifice, its altered core, its glaciers or the shallow units exposed around it.

rainier3d connects these scales in one model with three properties:

- it is tied to the mapped geology at the surface;
- it is consistent with regional tomography at depth;
- it reproduces the travel times of the Pacific Northwest Seismic Network (PNSN).

The same grids carry the surface layers that describe soil, water, vegetation and ice, the geodetic strain field, and the stress of the edifice load. The model is a digital model rather than a digital twin: it assimilates no time-dependent data. Only the GNSS product is refreshed on a schedule.

This paper describes, in order:

2 Domain, grids and workflow

The model uses Universal Transverse Mercator zone 10N (EPSG:32610) coordinates, with elevation in metres above NAVD88 (EPSG:5703), positive up.

Table 2. Model grids. The master product is one xarray DataTree in Zarr v3 format (`model.zarr`) with one node per level and a surface node.

Level	Elevation range (m)	Horizontal spacing (m)	Vertical spacing (m)	Cells ($z \times y \times x$)
Surface grid	–	100	–	750×700
L1	4400 to 0	250	50	$88 \times 300 \times 280$
L2	0 to –6000	500	250	$24 \times 150 \times 140$
L3	–6000 to –20000	1000	1000	$14 \times 75 \times 70$

- 40 – **Model box.** Easting 553–623 km and northing 5149–5224 km: the geographic box 122.30–121.40° W, 46.50–47.15° N, projected and rounded outward to whole kilometres.
- **Summit.** Columbia Crest (4392 m) lies at easting 594.5 km, northing 5189.5 km.
- **Single source of the geometry.** All grids are derived from one configuration file (`configs/domain.yaml`), and the code never hard-codes bounds, spacing or coordinate system.

45 Properties are stored on three stacked regular grids whose spacing coarsens with depth (Table 2). Cells above the ground carry no properties. The depth of each cell below the local ground surface is stored with it.

 The workflow is a sequence of numbered scripts, S0 to S23, each run as a task of one locked software environment (pixi). Fig. 1 shows how they connect:

- the model chain (S1–S5);
- 50 – the calibration loop, in which travel-time residuals update the rock-physics parameters and the regional correction (S13, S14);
- the checks and products that read the finished model.

 The domain and its sensors are mapped in Fig. 2.

3 A deterministic data compilation

55 3.1 Principles

 The database is compiled by code, not by hand, following five rules.

1. **One registry.** `configs/sources.yaml` lists every data set and publication the model uses (89 entries). Each entry gives its DOI or service address, licence, the date and method of verification, and its role in the model. Every number in the configuration files names a registry key. A value chosen by the authors carries the key `m1_placeholder`, so the unsourced values can be listed and replaced. The bibliography of this paper is generated from the same registry (`pixi run bib`), with BibTeX keys equal to the registry keys.

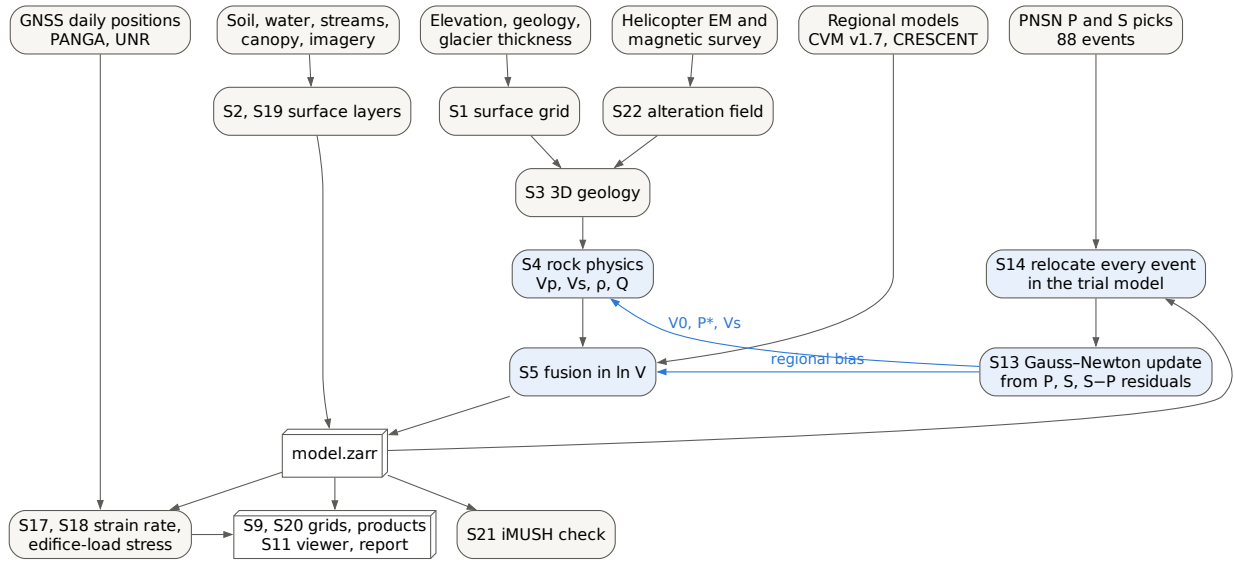


Figure 1. Workflow. Inputs (left) are fetched from their original archives. The model chain (middle) builds the surface grid, the 3D geology, the rock physics and the fusion with the regional models. The calibration (blue) relocates every PNSN event in each trial model and updates the rock-physics multipliers (S4) and a static correction of the regional models (S5) from the P, S and S–P residuals. Checks and products read the finished model (right).

2. **Original archives, cached.** Each stage downloads what it needs from the original archive, clipped to the model box where the service allows it. Examples are a window of a cloud-optimised GeoTIFF, an OPeNDAP subset, a feature-service query or a staged file. Each download is cached under `data/raw/<source>/`, and reruns read the cache.
3. **A checksum for every cached file.** `scripts/00_data_manifest.py` records the size and SHA-256 of all cached files in the committed table `docs/data_manifest.csv` (2,072 files, 4.44 GB; Table 3). `pixi run manifest -- --check` compares a rebuilt cache with it. Some services can return different bytes on a later request: ComCat can revise picks, and the USGS can re-stage NHDPlus. The check lists those files, so any difference between a rebuilt model and the published one can be traced to its input.
4. **One environment.** The software is pinned in `pixi.lock` for Linux and macOS (arm64). Two slim environments serve automation: `gnss` for the weekly strain refresh and `paper` for this report.
5. **Invariants.** The test suite checks the built model against rules that must hold whatever the data:

- no properties above the ground;
- V_p/V_s and density within physical bounds;

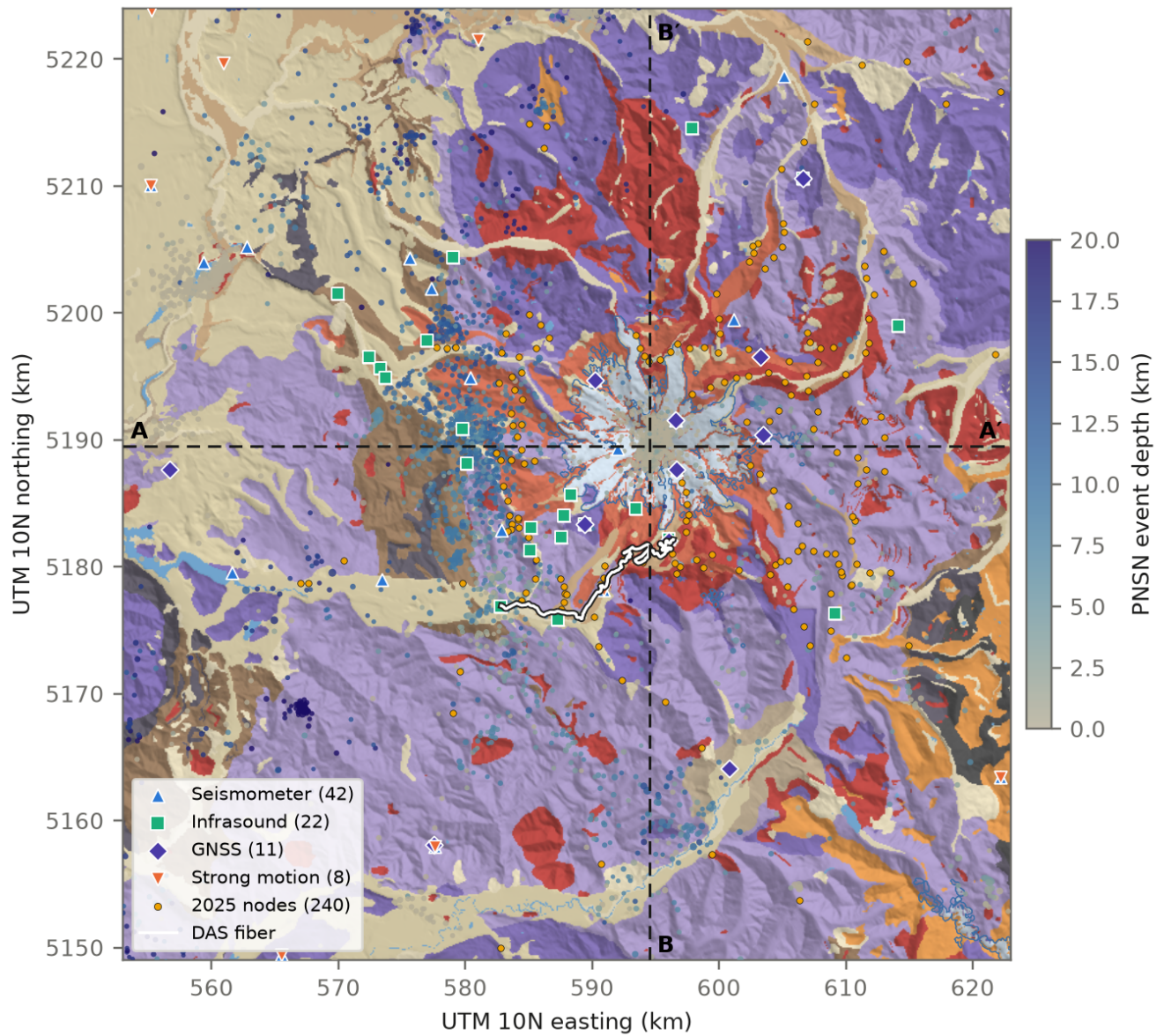


Figure 2. Model domain. The background shows the surface model units derived from the Washington 1:100,000 geologic map over a hillshade, with glacier outlines in blue. Symbols are operating sensors (FDSN metadata, GNSS site metadata and a 2025 nodal deployment). The white line is the Paradise–Nisqually Entrance distributed acoustic sensing (DAS) fibre, and dots are PNSN earthquakes since 2015 ($M \geq 0.5$), coloured by depth. The dashed lines mark sections A–A' and B–B'.

- the long wavelengths of the fused model equal those of the regional model within a tolerance;
- the surface unit matches the unit of the top model cell.

Tolerances are fixed in the configuration and are not relaxed to make a run pass.

Table 3: The raw input cache, from `docs/data_manifest.csv`. SOLUS100 soil thickness is read directly from its cloud-optimised GeoTIFFs and is not cached.

Source folder	Stage	Content	Files	Size (MB)
dem	S1	USGS 3DEP 1 arc-second DEM, clipped	1	59
geology	S1, S24	Washington 1:100,000 geology (GeMS) map units and faults; Quaternary faults	3	5
glaciers	S1	IceBoost v2 per-glacier thickness, RGI 6.0 outlines, GlaThiDa	222	44
ecology	S2	ETH canopy height 2020, NLCD 2021, clipped	2	27
hydrology	S2	NHDPlus HR geodatabases (HU4 1703, 1708, 1711), two water-table grids	7	924
imagery	S2	Sentinel-2 L2A median composite	1	96
regional	S5	Cascadia velocity model v1.7, clipped	1	60
pnsn	S6, S13	ComCat origins and phase picks (QuakeML)	93	21
sensors	S8	FDSN station and channel metadata, GNSS sites	2	1
gnss	S17	PANGA and UNR daily positions, velocity fields, manifest	80	218
emc	S21	iMUSH tomography (EMC netCDF)	1	5
rainier_aerogeophysics	S22	1996 helicopter EM and magnetic grids	8	19
wgs_landslides	S24	Washington landslide inventory: lidar-protocol deposits, recent landslides, compilation	3	3
allstadt2017	S24	Seismically recorded mass movements, western United States (<code>Events.csv</code>)	1	0.03
usgs_rainier_hazards	S24	Lahar hazard zones of 1998 (shapefiles)	1	0.2
dem_3dep_1m	S24	3DEP 1 m windows around landslide polygons; 3DEP source footprints	1,639	1,818
canopy_storage	S28	GEDi L3 global 1 km grids and their crops to the box (vendored canopy-storage code)	7	1,135

Four inputs are not fetched by script:

- 80 – **Cascadia velocity model v1.7.** ScienceBase blocks scripted downloads with a captcha, so the files are downloaded by hand once.
- **CRESCENT Gen0.** The model file is read from a local copy of its figshare record (He, 2026).
- **PNSN one-dimensional model.** The table is read from a local file.
- **Canopy-storage products.** The lidar canopy and soil-map products are delivered files (Sect. 4.3).

85 The code names each of these paths, and their registry entries say how to obtain them.

3.2 Rebuilding the database and the model

From a clean clone, the following commands download every input, rebuild the model and check the result. Stages that need credentials or delivered files skip those layers and say so.

```
git clone https://github.com/Denolle-Lab/mt-rainier-digital-model && cd mt-rainier-digital-model
pixi install                                # the locked environment
pixi run all                               # S1-S8: surface, layers, geology, rock physics, fusion, checks,
↳ figures
pixi run s2 -- --ma                        # optional: the Ma et al. (2026) water table (~1 GB download)
pixi run s22 && pixi run s3 && pixi run s4 && pixi run s5 # alteration from the EM survey, then
↳ rebuild
pixi run s17 && pixi run s18                # GNSS positions, velocities, strain, edifice load
pixi run s24                              # mass movements and faults: catalogue, figure, viewer layers (~1.8
↳ GB of 1 m windows)
pixi run -e canopy canopy                  # canopy-storage pipeline: GEDI (Earthdata login), Sentinel-2 LAI
↳ (CDSE client)
pixi run s9                              # uniform grids for ray tracing and location
pixi run manifest -- --check              # compare the rebuilt cache with docs/data_manifest.csv
pixi run test                             # unit tests and the invariants of the built model
```

The calibration (S13, about 26 min on a 10-core laptop) and the relocation comparisons (S14) are rerun as described in
90 docs/joint_calibration.md.

3.3 Redistribution

The pipeline is published in full. Derived products are published unless a source forbids it.

- **Excluded products.** A variable whose provenance includes a source flagged `redistribute_derived: false` is left out of the downloadable archives.
- 95 – **Flagged sources.** Three sources carry the flag:
 - the Ma et al. (2026) water table, whose licence (CC-BY-NC-ND 4.0) forbids derivatives;
 - the lidar canopy products;

- the soil-map image, whose licences are not documented.

- **Rebuilding the excluded layers.** Users rebuild them with their own access. Two open archives need a free login to download, NASA Earthdata for GEDI and the Copernicus Data Space for the Sentinel-2 leaf area index. Their derived layers are published.

`docs/data_policy.md` gives the tier of every source.

4 Surface layers

4.1 Elevation, geology and glaciers

Three surface layers set the top of every model column.

- **Elevation.** USGS 3D Elevation Program at 30 m, block-averaged to the 100 m surface grid. It ranges from 24 to 4380 m in the box.
- **Geology.** The Washington Geological Survey 1:100,000 surface geology in the GeMS format (Washington Geological Survey, 2026a).

- All 149 map symbols present in the box are assigned to 14 surface model units by rules on the symbol, checked against each unit’s full name in the map’s Description of Map Units. Every cell receives a unit.
- The rules are in `configs/units.yaml`, and the generated crosswalk is committed for review.
- The park map of Fiske et al. (1963) is carried as a georeferenced image for display.

- **Faults.** The same map gives 134 fault traces in the box (220 km), 106 of them high-angle dip-slip faults and 46 concealed. The Washington Quaternary fault layer adds four features (Washington Geological Survey, 2026b): the Western Rainier and Goat Rocks seismic zones, drawn as geophysical lineaments, the Devils Dream reverse fault and an unnamed oblique reverse fault. Neither source gives a dip, and no fault section of the 2023 National Seismic Hazard Model lies in the box (Hatem et al., 2021); the nearest, the Olympia and Tacoma faults, are 17 and 28 km outside it. Faults are therefore mapped traces (Fig. 21a) and are not surfaces in the 3D model.

- **Glacier ice thickness.** IceBoost v2 per-glacier grids (Maffezzoli, 2025) for the 219 glaciers of the Randolph Glacier Inventory 6.0 (RGI Consortium, 2017) whose centroids fall in the box, 97.3 km² in all. The grids are area-averaged onto the 100 m grid, and each glacier is rescaled so that its volume matches the IceBoost total for it. This removes an 18% overestimate from counting partly glacier-covered cells as full. The total is 5.50 km³, of which 5.32 km³ is on the cone.

The glacier bed is the ground elevation minus the ice thickness. Against the 1981 radar surveys of Driedger and Kennard (1986) as archived in GlaThiDa (World Glacier Monitoring Service (WGMS), 2020), maximum thickness agrees for Carbon, Nisqually and Tahoma glaciers (Fig. 3). IceBoost is thicker on Emmons (273 vs 185 m) and Winthrop (237 vs 98 m) glaciers. Part of any difference reflects thinning since 1981 (Sisson et al., 2011).

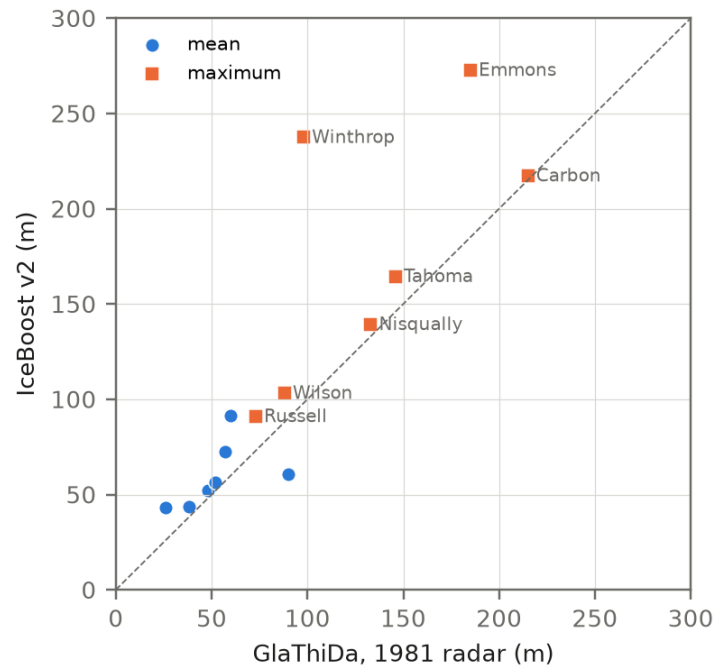


Figure 3. Mean (circles) and maximum (squares) ice thickness from IceBoost v2 against the 1981 ground-penetrating radar summaries in GlaThiDa, for seven Rainier glaciers. The dashed line is 1:1.

4.2 Soil, water, land cover and imagery

Script S2 reads each source over the model box only, resamples it onto the 100 m grid and records its source (Table 4, Fig. 4).

130 Continuous layers are averaged over each cell; categorical layers take the most common class.

Notes on individual layers:

- **SOLUS100.** It predicts soil depth only to 2.01 m, so that value is read as “at least 2 m”.
- **Sentinel-2 composite.** It uses the least-cloudy scenes (at most 25% cloud), up to six per Sentinel-2 tile, so that every part of the box is covered. The scene classification keeps vegetation, bare soil, water, snow and unclassified pixels, and the median is taken per pixel.
- **Reflectance offset.** Since processing baseline 04.00, Level-2A reflectance is stored with an offset of +1000 that the distribution services do not remove. The offset is removed before computing indices. Without the correction, median NDVI over this forest is 0.46 instead of 0.85.
- **Snow and ice.** NDSI above 0.4 covers 66 km² at the end of the 2025 melt season, against 97 km² of glacier in the Randolph inventory outlines (about 2000). The difference is consistent with debris-covered glacier tongues (Carbon, Emmons, Winthrop), which read as rock in NDSI, and with glacier retreat.

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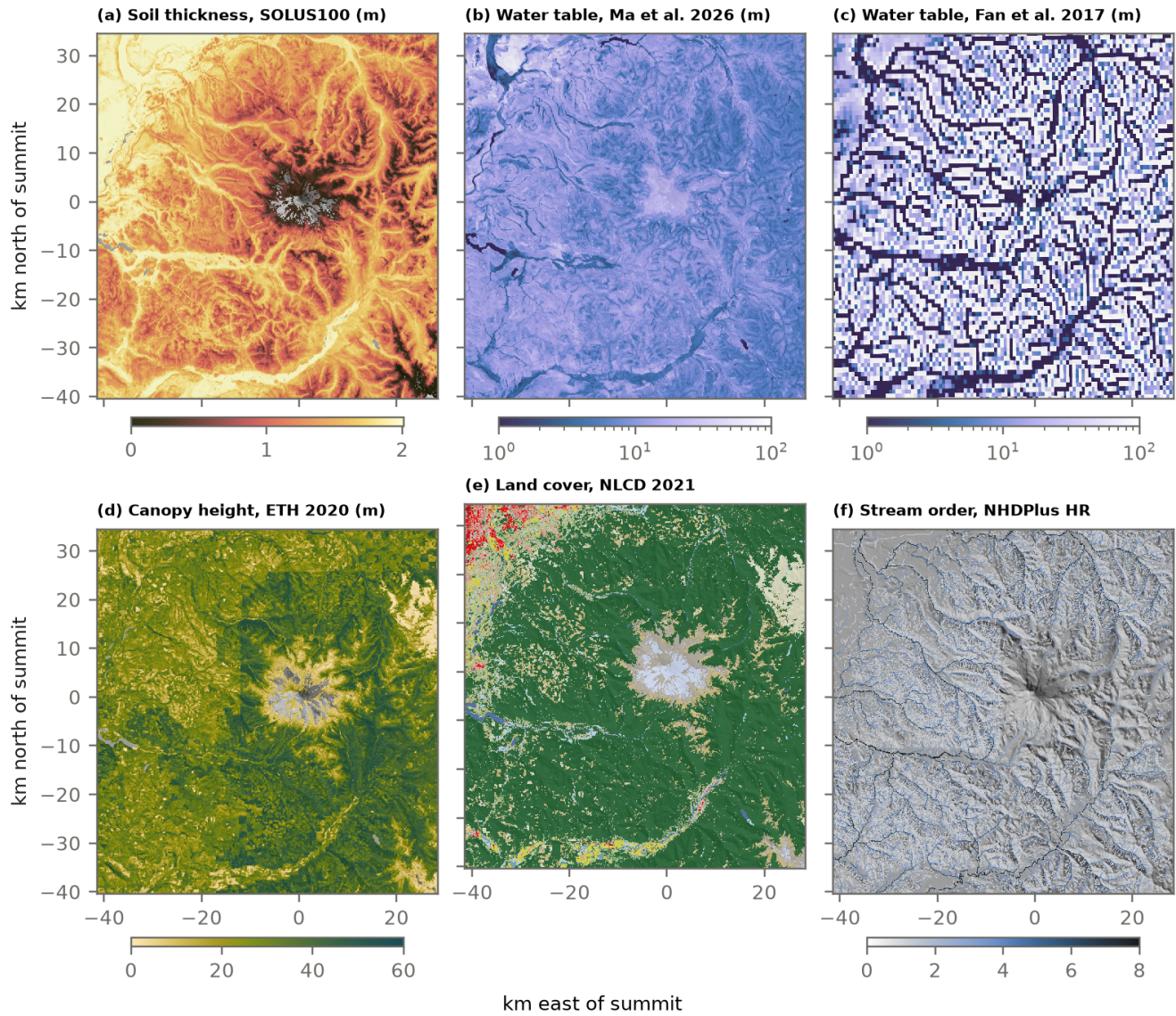


Figure 4. Environmental surface layers on the model grid, over a hillshade: (a) soil thickness, (b, c) water-table depth from two estimates on the same logarithmic scale, (d) canopy height, (e) land cover, (f) Strahler order of the NHDPlus HR flowlines. The rectangular step in (d) near 12 km west and 20 km north of the summit comes from the canopy product itself.

Table 4. Environmental surface layers (S2); percentiles are over the cells of the box.

Layer	Source	Native resolution	Median (5th–95th percentile)
Soil thickness	SOLUS100 depth to a lithic contact (Nauman et al., 2024)	100 m	1.41 m (0.54–1.97 m)
Water-table depth	Random-forest estimate for the conterminous US (Ma et al., 2026b)	about 24 m	10.1 m (4.8–20.4 m)
Water-table depth	Global groundwater model (Fan et al., 2017)	30'' (about 1 km)	23 m (0–306 m)
Streams	NHDPlus High Resolution, 60,174 flowlines, Strahler orders 1–8, with mean annual flow (U.S. Geological Survey, 2026b)	1:24,000	–
Canopy height	ETH global canopy height 2020 (Lang et al., 2023)	10 m	30 m (9–45 m)
Land cover	NLCD 2021 (U.S. Geological Survey, 2023)	30 m	evergreen forest in 72.5% of cells
NDVI	Sentinel-2 L2A median composite, 1 August – 30 September 2025 (European Space Agency, 2025)	20 m	0.85 (0.26–0.92)
NDSI	same composite	20 m	–0.48 (–0.60 to –0.24)

4.3 Vegetation structure

Script S19 adds the vegetation products of the canopy-storage project (M. Köpfli, University of Washington) on the same grid (Table 5, Fig. 5). They describe the canopy that intercepts precipitation and loads the ground: height, cover, leaf and plant area, and aboveground biomass.

The two canopy heights differ by design. The lidar value is the mean of 10 m cells in each 100 m cell, including gaps, while the GEDI value is a 1 km mean of the tallest return per footprint. That is why the lidar median (14.6 m) is lower than the GEDI median (25.5 m) and the ETH median (30 m).

The project’s download and gridding code is part of this repository, unmodified and under its MIT licence (`third_party/canopy-storage`). Script S28 runs it inside the raw-data cache: it fetches the GEDI L3 grids and L2B footprints through NASA Earthdata and the Sentinel-2 leaf area index through the Copernicus Data Space, and S19 reads its products in place of the delivered files. The GEDI L3 canopy height it produces is identical to the delivered grid (33,701 cells). The L2B plant area index, the L4B biomass, the lidar layers and the soil map are still read from the delivered files: the vendored gridder writes the maximum plant area index rather than the mean, and the others were made outside that code. The project’s lidar scripts compute canopy height only at its stations; the 10 m lidar grids were made from the same tiles in QGIS. `docs/canopy_pipeline.md` lists each service call.

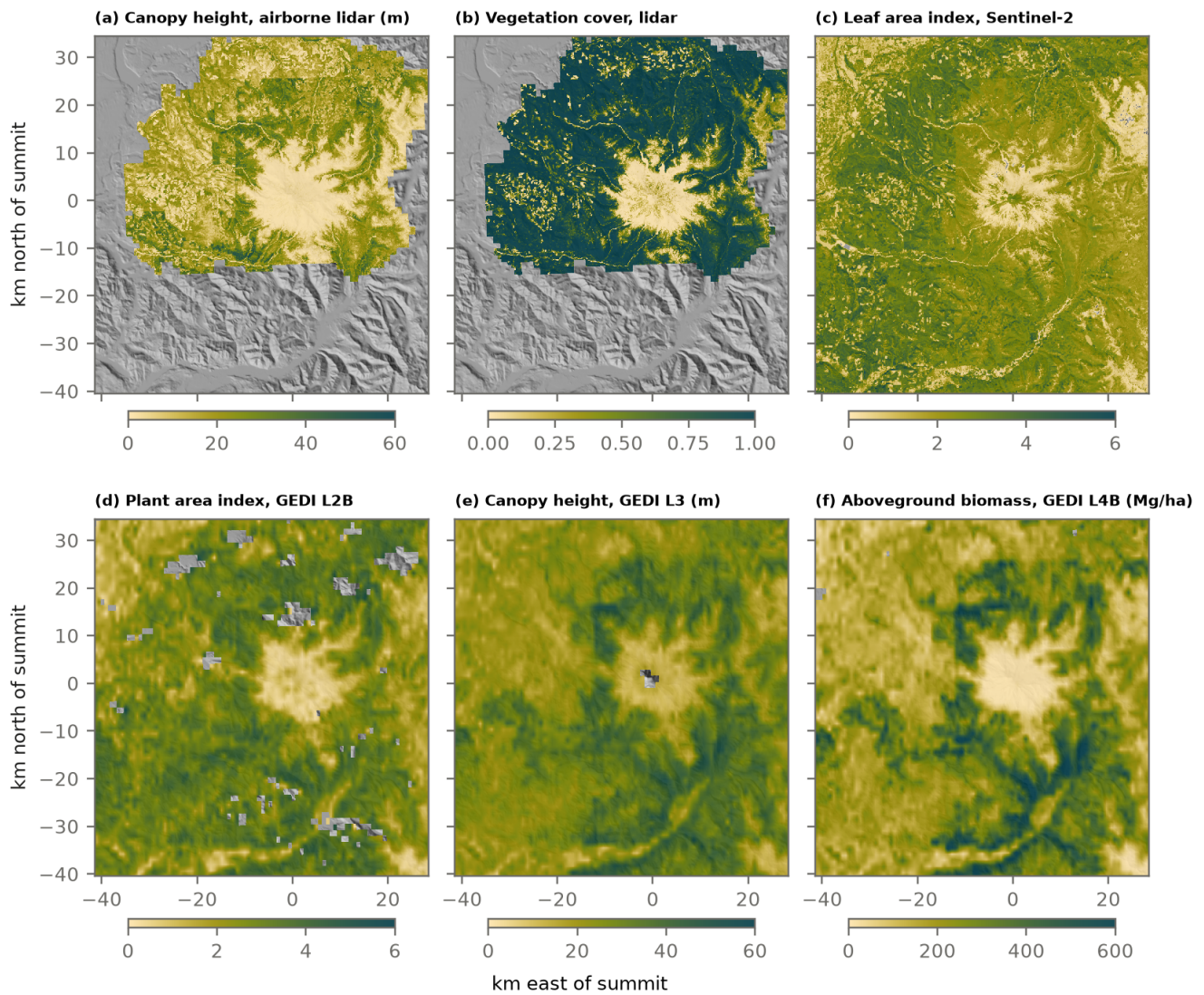


Figure 5. Vegetation layers on the model grid, over a hillshade: (a) lidar canopy height and (b) vegetation cover, (c) Sentinel-2 leaf area index, (d) GEDI plant area index, (e) GEDI canopy height, (f) GEDI aboveground biomass. Straight edges in (a) and (b) are lidar tile boundaries.

Table 5. Vegetation layers (S19). The lidar is the 2022–2023 “Wali” acquisition of the Washington Department of Natural Resources Lidar Portal, whose terms of reuse are not yet confirmed; the provenance of a soil-map image delivered with these products is undocumented. Both are displayed in the 3D viewer and excluded from the downloadable products.

Layer	Source	Native resolution	Median (5th–95th percentile)
Canopy height	airborne lidar, digital surface minus terrain model	10 m	14.6 m (0.3–32.4 m), on 49% of the box
Vegetation cover	same lidar	10 m	0.88 (0.01–1.00), on 49% of the box
Leaf area index	Sentinel-2 L2A, SNAP biophysical processor, 1 July – 15 August 2023 (European Space Agency, 2023)	10 m	2.36 (0.40–3.68)
Plant area index	GEDI L2B, footprints gridded (Dubayah et al., 2021b)	1 km	2.72 (0.86–3.79)
Canopy height	GEDI L3, mean relative height 100 (Dubayah et al., 2021a)	1 km	25.5 m (13.3–39.9 m)
Aboveground biomass	GEDI L4B, with standard error (Dubayah et al., 2025)	1 km	198 Mg ha ^{−1} (58–436)

Table 6. Column rules for the 3D units (S3; `configs/units.yaml`).

Condition	Assigned unit
$d < 0$	air
$d < \text{ice thickness}$	ice
$d < \text{ice} + \text{deposit thickness}$	mapped surface deposit
inside the edifice footprint, above the edifice base	Rainier andesite
young volcanic rocks, or andesite outside the footprint, within 300 m of the bedrock top	that cap unit
above the base of the bedrock unit	bedrock unit (mapped, or nearest mapped basement unit)
otherwise	middle crust

5 Subsurface model

5.1 Three-dimensional geology

Units are extended to depth by explicit column rules, in the manner of the San Francisco Bay region model (Aagaard and Hirakawa, 2021): the geology comes first and the velocity rules second. For a cell at elevation z and depth d below the ground, the rules of Table 6 are applied from the top down.

- **Edifice.** The edifice footprint is the andesite and ice mapped within 12 km of the summit, closed over 500 m. Its base is the pre-volcanic surface: the elevations where basement rocks crop out around the footprint, interpolated linearly across it. The base lies at 804–2234 m over 252 km².

165 – **Basement.** Beneath the deposits and the edifice, the basement unit is taken from the nearest outcrop.

- The supracrustal units (the Ohanapecosh, Fifes Peak and Stevens Ridge formations, the Eocene volcanic rocks and the Puget Group) extend to 4 km below sea level.
- The Miocene plutons (Tatoosh, White River, Carbon River) extend to 10 km below sea level.
- The Mashel Formation is capped at 300 m thickness.

170 – **Unconsolidated deposits.** They have fixed thicknesses: glacial drift 15 m, alluvium and colluvium 10 m, lahar deposits 15 m.

- **Magma body.** A slow body below the summit is an ellipsoid centred 11.5 km below sea level, with semi-axes of 5 km horizontally and 6.5 km vertically. It follows the low-Vp body imaged 5–18 km beneath the summit by Moran et al. (1999).

175 5.2 Hydrothermal alteration from the helicopter electromagnetic survey

Clay-bearing altered rock is electrically conductive, while fresh lava and ice are resistive. The 1996 helicopter electromagnetic (EM) and magnetic survey of Rainier (Rystrom et al., 2000) measured apparent resistivity at four frequencies (33 kHz, 4737 Hz, 4341 Hz and 837 Hz) on a 50 m grid. Finn et al. (2001) used it to map collapse-prone altered zones. Script S22 turns these grids into an alteration field in three steps.

- 180 1. **Registration.** The grids are converted from NAD27 to the model datum. Their orientation is checked against the data, not assumed: the magnetic anomaly correlates with high-passed topography at 0.60 as read, and at 0.08 if flipped north–south.
2. **Intensity.** The survey saturates at a different resistivity at each frequency, so each frequency f is referred to its own fresh level L_f . L_f is the median log resistivity over the surveyed edifice lavas: 4.35, 3.89, 3.74 and 3.00 ($\log_{10} \Omega \text{ m}$). The intensity is
- 185

$$a_f = \text{clip}\left(\frac{L_f - 0.3 - \log_{10} \rho_a}{0.7}, 0, 1\right), \quad (1)$$

which starts at half the fresh resistivity and reaches 1 at a tenth of it.

3. **Depth.** Each frequency senses to a depth of investigation of half its skin depth, $503\sqrt{\rho_a/f}$ m, capped at 150 m. The three-dimensional field is the maximum over frequencies of a_f tapered below that depth. Depth is measured below the glacier bed, and the field is restricted to the edifice lavas and young volcanic rocks.
- 190

Over the 126 km² of surveyed edifice lava, 10.6 km² reaches an intensity of 0.5 or more (Fig. 6). That corresponds to 1.9 km³ of equivalent fully altered rock in the model grid. Table 7 gives the distribution around the summit.

Table 7. Alteration of the surface rock from the EM survey (S22, docs/alteration.md).

Area (edifice lavas within 6 km of the summit)	Mean intensity at the surface	Share with intensity ≥ 0.5
West flank	0.08	8%
Summit (within 1.5 km)	0.21	19%
East flank	0.27	25%

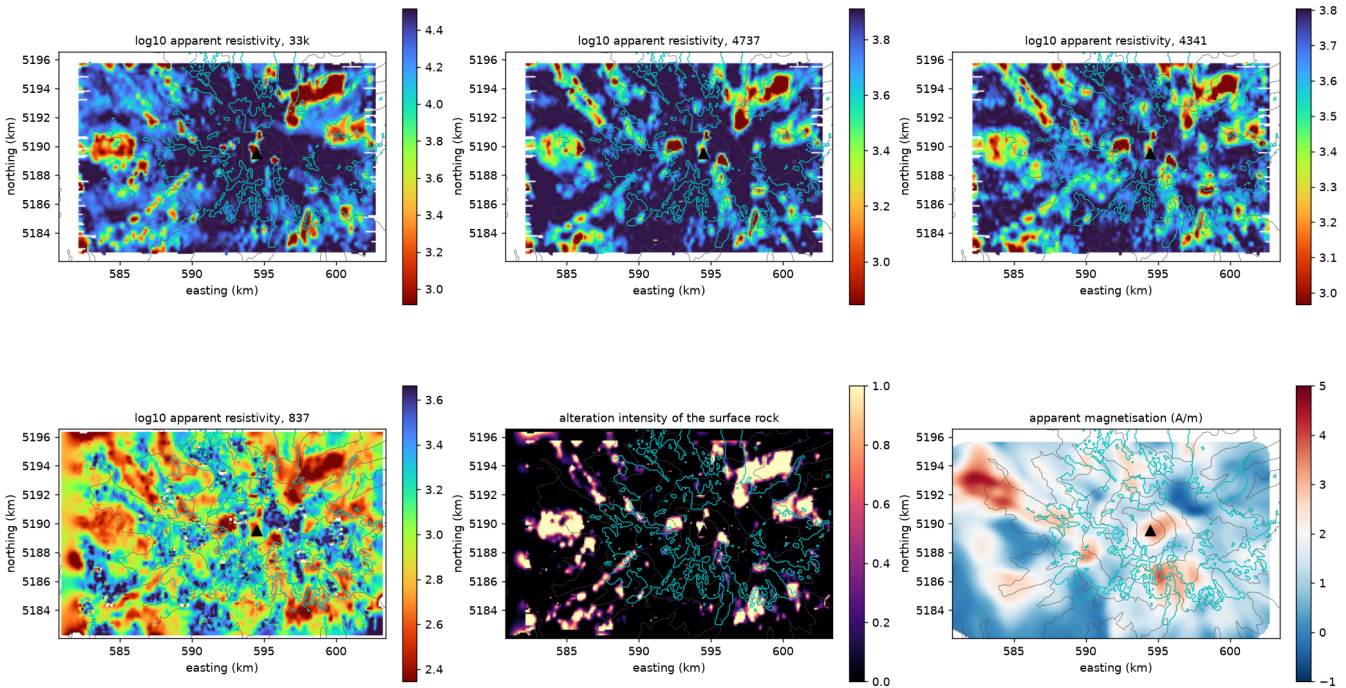


Figure 6. Apparent resistivity at the four EM frequencies, alteration intensity of the surface rock and terrain-correlated apparent magnetisation (S22). Contours: elevation every 500 m; cyan: glaciers thicker than 10 m; triangle: summit.

The concentration east of and around the summit agrees with the exposed east–west belt of altered rock described by Finn et al. (2001) and John et al. (2008). The EM field sees only the top ~150 m. It cannot see the altered rock buried beneath
195 fresh cover on the upper west flank, which Finn et al. (2001) inferred from magnetic modelling. The survey’s terrain-correlated apparent magnetisation is published as a separate map.

5.3 Rock physics

Each rock unit receives a P-wave speed that increases with effective pressure as cracks close:

$$V_P(P) = V_\infty - (V_\infty - V_0) e^{-P/P^*}, \quad P = (\rho_b - \rho_w) g d, \quad (2)$$

200 with bulk density $\rho_b = 2500 \text{ kg m}^{-3}$, water density $\rho_w = 1000 \text{ kg m}^{-3}$ (hydrostatic pore pressure) and d the depth below the local ground.

- **Vp/Vs and density.** Where a unit has no ratio or density of its own, both follow Brocher (2005): the regression of Vs on Vp, and the Nafe–Drake density relation (valid for $1.5 < V_p < 8.5 \text{ km s}^{-1}$).
- **Attenuation.** $Q_S = 0.05 V_S$ (Vs in m s^{-1}) and $Q_P = 2 Q_S$.
- 205 – **Parameters.** Table 8 lists the unit parameters, chosen within published ranges. The calibration of Sect. 5.5 scales them with three global multipliers. The contrasts between units therefore come from the table, and their overall level comes from the travel times.
- **Magma body.** Inside it, Vp, Vs and density are reduced by 10%, 15% and 3%.
- **Alteration.** The alteration intensity a scales Vp by $(1 - 0.30a)$, Vs by $(1 - 0.35a)$ and density by $(1 - 0.12a)$, within
- 210 the ranges reported for altered volcanic rock (Heap and Violay, 2021).

5.4 Fusion with the regional models

Regional models. The regional model has two parts:

- **Down to 9.9 km below the ground:** the USGS Cascadia velocity model v1.7 (Wirth et al., 2025), for Vp and Vs. Its depth axis is below the ground surface. For the shallow level this is confirmed by a median top-sample Vs of 206 m s^{-1} ,
- 215 and for the deeper level it is inferred from continuity.
- **Below 9.9 km:** CRESCENT Gen0 Vs (He et al., 2026), with Vp from Brocher (2005).

Below 1 km, both are multiplied by a static, depth-dependent correction fitted to the PNSN arrivals (Sect. 5.5). The corrected Vs is floored at $V_p/1.6$ (Christensen, 1996).

220 **Strategy.** Neither model is right at every wavelength. The regional models resolve the crust at wavelengths of several kilometres and are constrained by teleseismic, ambient-noise and local-earthquake data; they carry no information on the mapped units. The geology model carries the mapped units and their contrasts, but its absolute level rests on laboratory-style rules. The fusion keeps each where it is informative: the regional model supplies the long wavelengths and the geology model the short ones. The combination is done in the logarithm of velocity, so contrasts are relative:

$$\ln V = \text{LP}_{\lambda_c}(\ln V_{\text{reg}}) + [\ln V_{\text{geo}} - \text{LP}_{\lambda_c}(\ln V_{\text{geo}})], \quad (3)$$

225 where LP is a horizontal Gaussian low-pass filter with half power at the cutoff wavelength λ_c , that is $\sigma = \sqrt{2 \ln 2} \lambda_c / 2\pi$.

Implementation details.

Table 8. Unit parameters of the crack-closure law (`configs/petrophysics.csv`). For the rock units (Rainier andesite to middle crust, and the magma body), the calibration multiplies V_0 by 1.31 (capped at $0.98 V_\infty$), P^* by 0.91 and V_s by 0.976. Ice and the unconsolidated deposits keep their table values.

Unit	V_0 (km s ⁻¹)	V_∞ (km s ⁻¹)	P^* (MPa)	V_p/V_s	Density (g cm ⁻³)
Ice	3.80	3.80	–	1.98	0.917
Glacial drift	1.60	2.40	5	2.50	1.95
Alluvium, colluvium	1.50	2.20	5	2.80	1.90
Lahar deposits	1.80	2.80	8	2.30	2.00
Rainier andesite	2.80	5.60	30	Brocher	Nafe–Drake
Young volcanic rocks	3.00	5.60	30	Brocher	Nafe–Drake
Miocene intrusive rocks	4.50	6.20	40	Brocher	Nafe–Drake
Miocene volcanic rocks	3.20	5.80	40	Brocher	Nafe–Drake
Ohanapecosh Formation	3.40	5.90	40	Brocher	Nafe–Drake
Eocene volcanic rocks	3.50	6.00	40	Brocher	Nafe–Drake
Puget Group, Eocene sedimentary rocks	2.60	5.20	40	Brocher	Nafe–Drake
Mashel Formation	2.20	4.50	30	Brocher	Nafe–Drake
Russell Ranch Formation	4.00	6.00	40	Brocher	Nafe–Drake
Middle crust	6.00	6.50	50	Brocher	Nafe–Drake

Table 9. Departure of the calibrated fused model from the corrected regional model at the regional wavelengths below 1 km depth, and mean offset over all cells (`outputs/fusion_report.csv`). The tolerance is 0.03.

Level	RMS of LP(ln V) –			
	LP(ln V _{reg}), Vp	Same, Vs	Mean ln(V / V _{reg}), Vp	Same, Vs
L1	0.011	0.012	+0.046	+0.077
L2	0.002	0.002	+0.001	+0.001
L3	0.0006	0.0006	0.0002	0.0001

1. **Fused quantities.** Vs and Vp/Vs are fused rather than Vp and Vs separately, so the fused Vp/Vs always lies between the two input ratios.
2. **Cutoff wavelength.** λ_c grows with depth below the ground: 6 km at the surface, 10 km at 2 km, and 20 km from 10 km down. The values approximate the lateral resolution of the regional models.
3. **Air cells.** Cells above the ground are excluded from the filters by normalised convolution.
4. **Overshoot.** A Gaussian high-pass overshoots at sharp contrasts and puts fast rims around slow bodies. Six alternating projections remove them: each cell is clamped between its geology and regional values, then the regional low-pass is restored.
5. **Top kilometre.** The top 300 m keep the geology model unchanged, and a linear taper reaches the fused model at 1 km. The regional models do not resolve this depth range; their shallowest layer is a generic near-surface model.
6. **Density.** Density follows Vp through the ratio of Nafe–Drake densities, which preserves the density contrasts between units.

Fig. 7 shows the three models along a section through the summit. The fusion is checked by an invariant (Table 9): below 1 km depth, the low-passed fused model must equal the low-passed regional model within an RMS of 0.03 in ln V.

5.5 Calibration on P and S–P travel times

Data. The data are the analyst picks of the 88 PNSN earthquakes with $M \geq 2$ in the box between 2015 and 2026 that have at least six picks, four of them P. They comprise 1823 P and 1280 S picks at 50 stations, with origins and picks from the USGS ComCat catalogue (U.S. Geological Survey, 2026a). The event list is committed (`configs/validation_events.csv`), so every run uses the same set. Picks are weighted by the RMS of PNSN’s own location residuals: 0.14 s for P and 0.23 s for S.

Travel times. They are computed on a 500 m grid with pykonal’s point-source solver (White et al., 2020), one solve per station and phase by reciprocity.

- **Solver accuracy.** Against analytic travel times on this grid, its RMS error is 8 ms in a uniform model and 11 ms with a velocity gradient; fteikpy gives 22 and 46 ms.
- **Topography.** Cells more than one cell above the ground carry the speed of sound in air, so rays follow the rock and cannot cut across valleys. The one-cell skin keeps every station in rock.

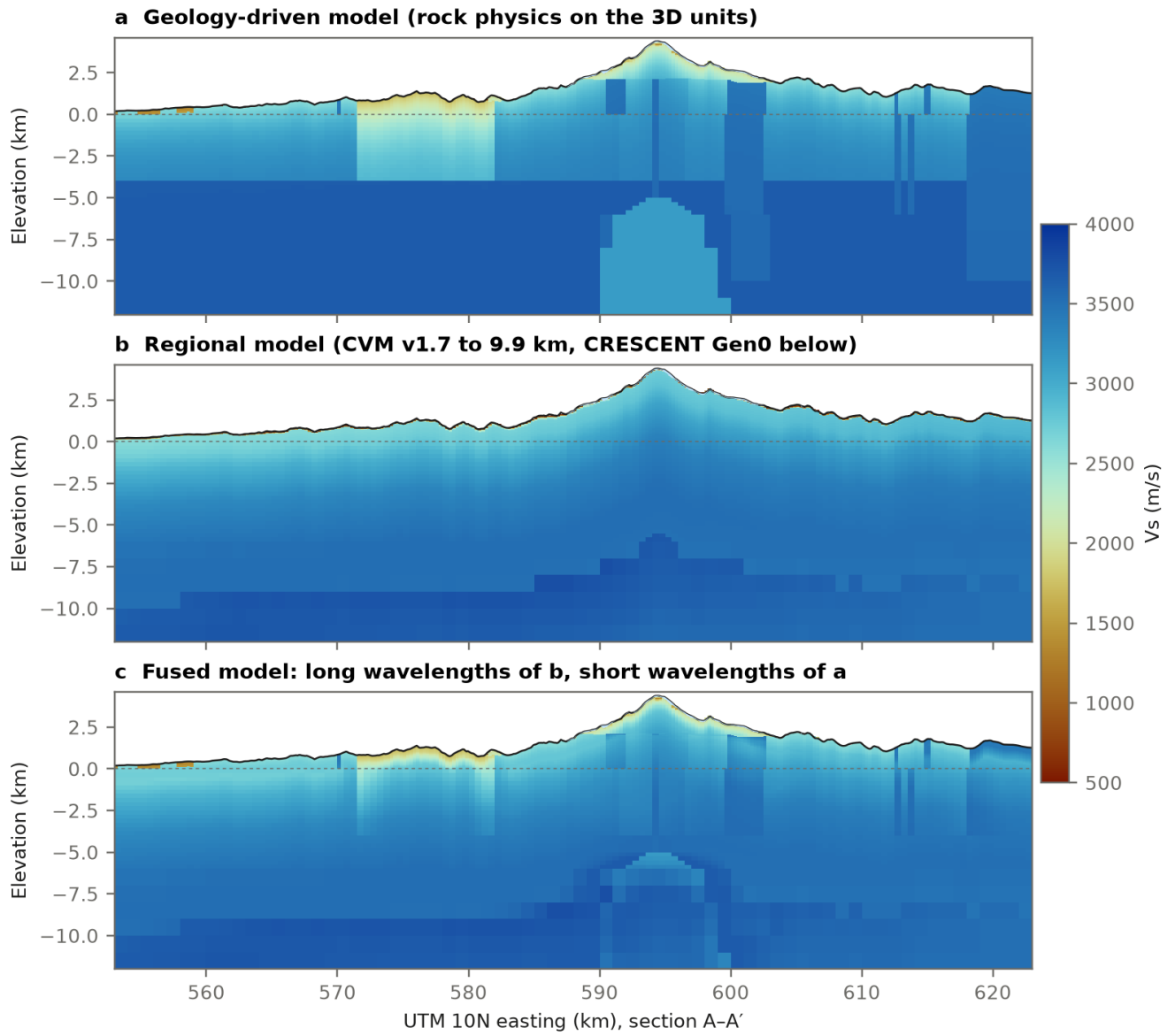


Figure 7. Vs along section A–A' (west–east through the summit): (a) geology model, (b) regional model, (c) fused model. The fused model keeps the regional long wavelengths and the geological contrasts: the Puget Group sedimentary block near 572–582 km, the Tatoosh and White River plutons, and the slow edifice.

- **Cross-check.** For station OBSR, NonLinLoc’s Grid2Time P times on the exported grids agree with the pykonal times to 12 ms on average (RMS 31 ms, 85 events).

What is estimated. Fifteen parameters θ scale the two models that are fused, and the hypocentres and origin times of the 88 events are re-estimated in every model tried. The travel-time data therefore constrain the velocity parameters only through the part of the misfit that relocation cannot absorb. This avoids the circularity of scoring a model with catalogue hypocentres that were located in a 1D model with station corrections.

Parameters.

- **Rock physics (3).** Log-multipliers on three quantities shared by all rock units: the zero-pressure velocity V_0 and the crack-closure pressure P^* of Eq. 2, and V_s . The V_s multiplier acts at fixed V_p , so it is a change of V_p/V_s . Ice and unconsolidated deposits are not scaled.
- **Regional correction (12).** A static depth profile $m(d)$ of the log-factor that multiplies the regional velocity, $V_{\text{reg}} \rightarrow V_{\text{reg}} e^{m(d)}$, one profile for V_p and one for V_s . Here d is depth below the ground. Each profile is piecewise linear between knots at 0, 1, 2, 4, 7, 11, 16 and 25 km and constant below 25 km. The values at 0 and 1 km are fixed at zero, because the fused model takes the top kilometre from the geology model (Sect. 5.4); the six deeper values are free.

Forward problem. For a trial θ , the model is rebuilt from its inputs by the same code that builds the published model. At $\theta = 0$ it returns the uncalibrated model cell for cell, which checks that the calibration fits the model that is delivered and not an approximation of it. The steps are:

1. V_p of every rock-unit cell from Eq. 2 with the scaled V_0 and P^* , and V_s from that V_p through the unit’s V_p/V_s ratio, or the regression of Brocher (2005) where the unit has none, times the V_s multiplier;
2. the regional V_p and V_s multiplied by $e^{m(d)}$;
3. the fusion of Eq. 3, including the geology-only top 300 m and the taper to 1 km;
4. P and S travel times from every station on the 500 m grid, with air above the ground, as described under Travel times;
5. every event relocated in that model (see Relocation below).

Objective. With t_{ij} the pick of phase at station j for event i , T_j the travel-time field, and $(\mathbf{x}_i, \tau_i)$ the hypocentre and origin time, the normalised residual is

$$r_{ij} = \frac{t_{ij} - \tau_i - T_j(\mathbf{x}_i; \theta)}{\sigma_{\text{ph}}}, \quad (4)$$

where σ_{ph} is the pick uncertainty of the pick’s phase: $\sigma_P = 0.14$ s for P picks and $\sigma_S = 0.23$ s for S picks. The calibration minimises

$$\Phi(\theta) = \sum_{ij} w_{ij} r_{ij}^2 + \theta_g^T \mathbf{C}_g^{-1} \theta_g + \lambda_s \|\mathbf{D}\mathbf{m}\|^2 + \lambda_d \|\mathbf{m}\|^2, \quad (5)$$

where the hypocentres minimise the first term for each θ . The weights w_{ij} are the Huber weights of the relocation (threshold 1.5σ), so outlying picks count less. θ_g holds the three rock-physics log-multipliers, with prior standard deviations of 0.3, 0.7

and 0.1 on the diagonal of $\mathbf{C}_g$ (factors of 1.35, 2 and 1.1). $\mathbf{m}$ holds the regional log-factors at all eight knots, with the two fixed at zero, and $\mathbf{D}$ takes second differences along depth. The smoothing weight λ_s and the damping weight $\lambda_d = 0.01$ are relative: both are multiplied by the mean diagonal of the data term for the regional parameters.

Update. Each iteration linearises the residuals about the current model and hypocentres, $\mathbf{r}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) \approx \mathbf{r} - \mathbf{G} \delta\boldsymbol{\theta}$, with $\mathbf{G} = \partial\mathbf{T}/\partial\boldsymbol{\theta}$ scaled like the residuals.

– **Separating hypocentres from velocity.** For each event, the columns $\mathbf{H}_i$ of partials with respect to $(\mathbf{x}_i, \tau_i)$ are formed at its current location. The event’s rows of $\mathbf{G}$ and $\mathbf{r}$ are projected onto the orthogonal complement of $\mathbf{H}_i$, through a QR factorisation of $\mathbf{H}_i$ (Pavlis and Booker, 1980). What remains is the part of the misfit that no shift of that hypocentre can remove. Events with five or fewer usable picks keep at most one degree of freedom after the projection and are left out of the update.

– **Step.** With the projected $\tilde{\mathbf{G}}$, $\tilde{\mathbf{r}}$ and the regularisation matrix $\mathbf{L}$ of Eq. 5,

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \left(\tilde{\mathbf{G}}^T \tilde{\mathbf{G}} + \mathbf{L} \right)^{-1} \left(\tilde{\mathbf{G}}^T \tilde{\mathbf{r}} - \mathbf{L} \boldsymbol{\theta} \right). \quad (6)$$

– **Relocation.** Every event is then relocated in the updated model before the next linearisation, so velocity and hypocentres are updated in alternation, as in the minimum one-dimensional model of Kissling et al. (1994). Relocation is a grid search over nodes at or below the ground, within 15 km of the catalogue epicentre, on the sum of lrl with the median origin time, followed by Huber Gauss–Newton (Huber, 1964) on interpolated travel times with a penalty on hypocentres more than 50 m above the ground. It follows the grid-search-then-refine design of NonLinLoc (Lomax et al., 2000).

Derivatives.

– **Rock physics.** A rock-physics multiplier changes every rock-unit cell, and its effect passes through the fusion filters, so its column of $\mathbf{G}$ is a one-sided finite difference ($h = 0.05$ in log) through the whole forward problem, with the hypocentres held fixed. That costs one full rebuild and eikonal solution per parameter, and it is recomputed at every iteration on the fitting half.

– **Regional correction.** For knot k the derivative is the ray integral

$$\frac{\partial T}{\partial m_k} = - \int_{\text{ray}} \beta(d) \phi_k(d) s \, d\ell, \quad (7)$$

where s is slowness, ϕ_k the linear interpolation weight of knot k at depth d , and $\beta(d)$ the share of the regional model in the fused model: 0 above 300 m, rising linearly to 1 at 1 km. Rays are traced down the gradient of each station’s travel-time field (Thurber, 1983). The integral assumes that a depth-only factor on the regional model passes through the fusion unchanged. The low-pass filter of Eq. 3 passes a factor that varies slowly across the box, but the clamping of fused values between the two input models is not linear, and is the likely cause of the difference below. Against a finite difference through the full forward problem, for the Vs knot at 4 km over all 1280 S picks, the ray derivatives correlate at 0.96 with a slope of 0.80: they underestimate the sensitivity by about 20%. The residuals are always evaluated with

Table 10. Gauss–Newton iterations of the calibration (S13, outputs/joint_calibration/history.json): RMS after relocation and the geology parameters. Most of the misfit reduction happens in the first iteration.

Iteration	P, fitting (s)	P, held out (s)	S, fitting (s)	S, held out (s)	$\ln V_0$	$\ln P^*$	$\ln V_s$
0 (uncalibrated)	0.124	0.121	0.314	0.317	0	0	0
1	0.099	0.094	0.195	0.189	0.131	0.268	0.017
2	0.097	0.093	0.192	0.188	0.162	0.297	0.033
3	0.097	0.092	0.193	0.188	0.267	0.403	0.016
4	0.097	0.093	0.192	0.188	0.256	0.176	−0.003
final (all events)	0.095	0.092	0.190	0.188	0.270	−0.099	−0.024

the full forward problem, so this error does not bias the data fit. At convergence it acts, to first order, like regularising the regional correction about 20% more weakly than λ_s and λ_d state.

Validation and choice of smoothing. Events alternate between a fitting half and a held-out half in origin-time order, and held-out events are relocated in every iterate, so their misfit is an independent score. At the first iteration, λ_s is chosen from 0.01, 0.1, 1 and 10 by the normalised RMS of the held-out separated residuals predicted by the linearised step: 0.720, 0.719, 0.733 and 0.778. The adopted value is 0.1.

Iterations and uncertainties. Four iterations on the fitting half are followed by two on all events, which reuse the last rock-physics derivatives (Table 10); the run takes about 26 min on a 10-core laptop. Posterior standard deviations are the square roots of the diagonal of $(\tilde{\mathbf{G}}^T \tilde{\mathbf{G}} + \mathbf{L})^{-1}$ at the last iteration, multiplied by the mean reduced χ^2 of the two phases (0.46 for P, 0.68 for S). They are linearised. They do not include the error of the ray derivatives, or the trade-off with hypocentres beyond what the projection removes.

The fit is resolved as follows (Table 11, Table 12, Fig. 8, Fig. 9).

- **V_0 .** It is the parameter the data require: $\times 1.31 \pm 0.02$. The table rock is too slow near the surface. With the multiplier, surface V_p is 3.7 km s^{-1} for Rainier andesite and 4.5 km s^{-1} for the Ohanapecosh Formation. Because the multiplier is global, the Miocene plutons reach the cap of $0.98 V_\infty$ and are nearly uniform from the surface down.
- **P^* .** It is not resolved: its log-multiplier ranged from +0.40 to −0.10 across the iterations while the misfit changed by less than 1 ms, and it trades off against the regional correction at 2 km.
- **Rock V_s .** It falls by $2.4 \pm 1.2\%$.
- **Regional correction.** It changes V_p by less than 3.5% at every depth. It makes V_s 5–9% faster at 2–7 km and 5–8% slower at 16–25 km, which lowers V_p/V_s at 2–4 km depth from 1.83 to 1.74.

Table 11. Geology multipliers (`configs/velocity_calibration.yaml`, `block geology`). Posterior standard deviations are linearised and scaled by the reduced χ^2 .

Parameter	Multiplier	Posterior s.d. (ln)	Resolved
V_0 , zero-pressure V_p	1.31	0.018	yes
P^* , crack-closure pressure	0.91	0.098	no
V_s of rock units	0.976	0.012	marginally

Table 12. Static correction of the regional models (`block regional_bias`). Formal standard deviations of the log-factors are 0.003–0.016. The deepest knots rest on few rays: 30 events are deeper than 11 km and 8 deeper than 16 km.

Depth below ground (km)	2	4	7	11	16	25
Factor on regional V_p	1.022	1.005	0.968	0.975	0.979	0.973
Factor on regional V_s	1.049	1.086	1.060	0.990	0.952	0.925

- **Agreement between the two models.** After calibration, the geology model and the corrected regional model agree to within 3.5% between 0.3 and 4 km depth. Uncalibrated, they differ by up to 24%, and the fusion invariant exceeds its tolerance in L1 (0.032).

Why this parameterisation. Two simpler parameterisations were tested against the same data (Table 13).

- **V_s -only depth factor, catalogue hypocentres fixed.** It fits S as well as the adopted model, but pushes V_p/V_s to 1.60–1.67 at 2–4 km. That is low for crustal rock and lower still than expected beneath a volcano with a hydrothermal system: the S–P misfit of fixed hypocentres is partly a location error.
- **Depth factors on the regional V_p and V_s at all depths, with relocation.** It fits as well as the adopted model, but places the correction in the regional model where the fusion does not use it. It requires V_p 7–19% faster in the top 2 km, where the fused model follows the geology, and it breaks the fusion invariant (0.050).
- **Adopted parameterisation.** Fitting the geology’s rock physics and correcting the regional model only below 1 km fits equally well, satisfies the invariant (0.011 for V_p , 0.012 for V_s) and keeps the unit contrasts in physical parameters.

Alteration and the calibration. The calibration was run with a conduit-centred alteration field. Replacing it with the EM-based field of Sect. 5.2 changes the relocated RMS from 0.095 to 0.093 s for P and leaves S at 0.190 s, so the calibrated parameters are kept.

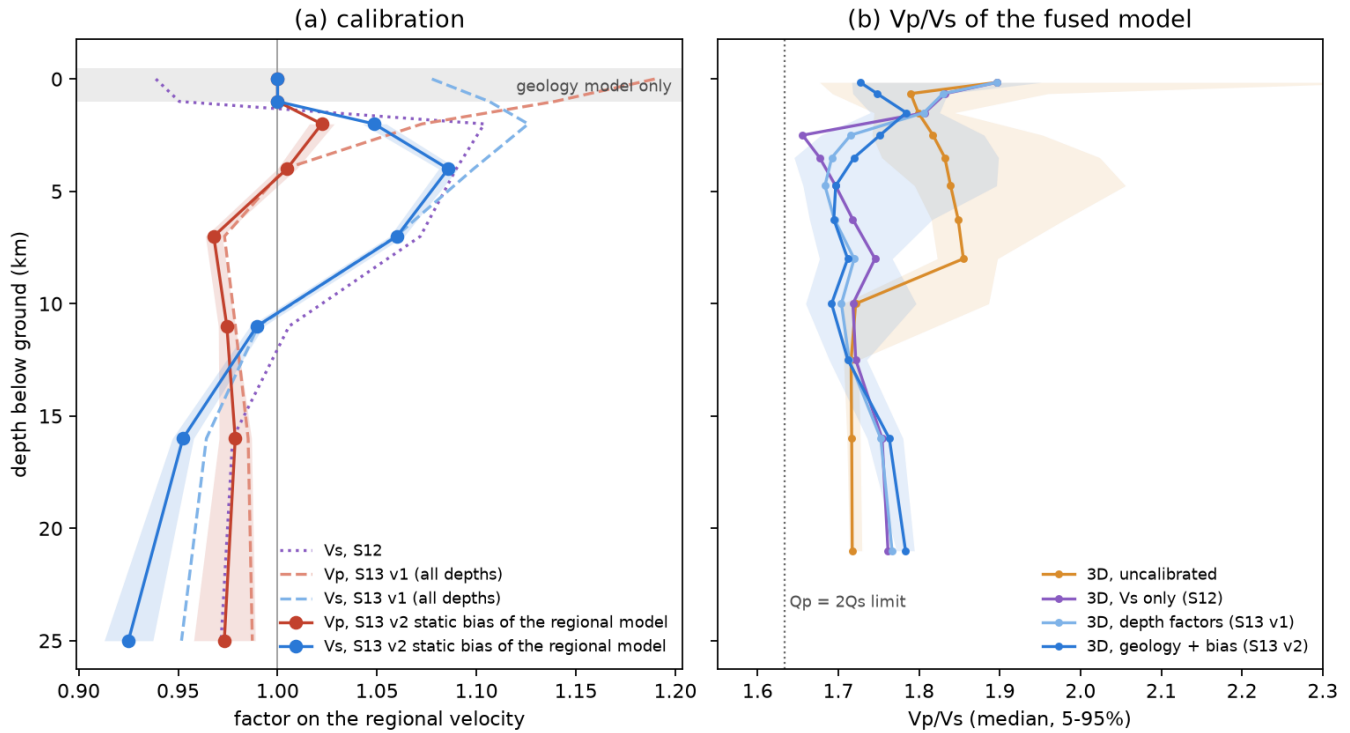


Figure 8. (a) Static correction of the regional Vp and Vs (solid, ± 1 standard deviation). The dashed and dotted lines are the two alternative parameterisations of Table 13. The grey band is the geology-only zone. (b) Median and 5–95% range of Vp/Vs of the fused model against depth below ground.

5.6 Validation

Held-out events. On the 44 events not used in the fit, relocated in each model, the RMS falls from 0.121 to 0.093 s for P and from 0.317 to 0.188 s for S. The fitting half ends at 0.097 and 0.192 s, so the fit does not overfit.

All models scored the same way. Table 13 relocates all 88 events in each model with the same locator and topography. Relocation alone does not rescue the uncalibrated three-dimensional model: its S residuals stay larger than those of the 1D model. The S–P misfit therefore lies in the velocity model, not in the catalogue hypocentres. With the ground bound, no event is placed at the top of the grid, and two of the 88 events end within 50 m above the ground.

Independent tomography. The iMUSH project imaged Vp and Vs around Mount St. Helens from local earthquakes and explosions recorded by a 70-station array (Ulberg et al., 2020; Incorporated Research Institutions for Seismology, 2020).

- **Coverage.** Its grid spans the whole model box. Its Vp is resolved over 82–90% of the box. Its Vs is resolved over 83–93% of the southern strip (south of 46.65° N) but only 11–19% of the northern half. Script S21 samples our models at its 51,300 resolved nodes below ground.

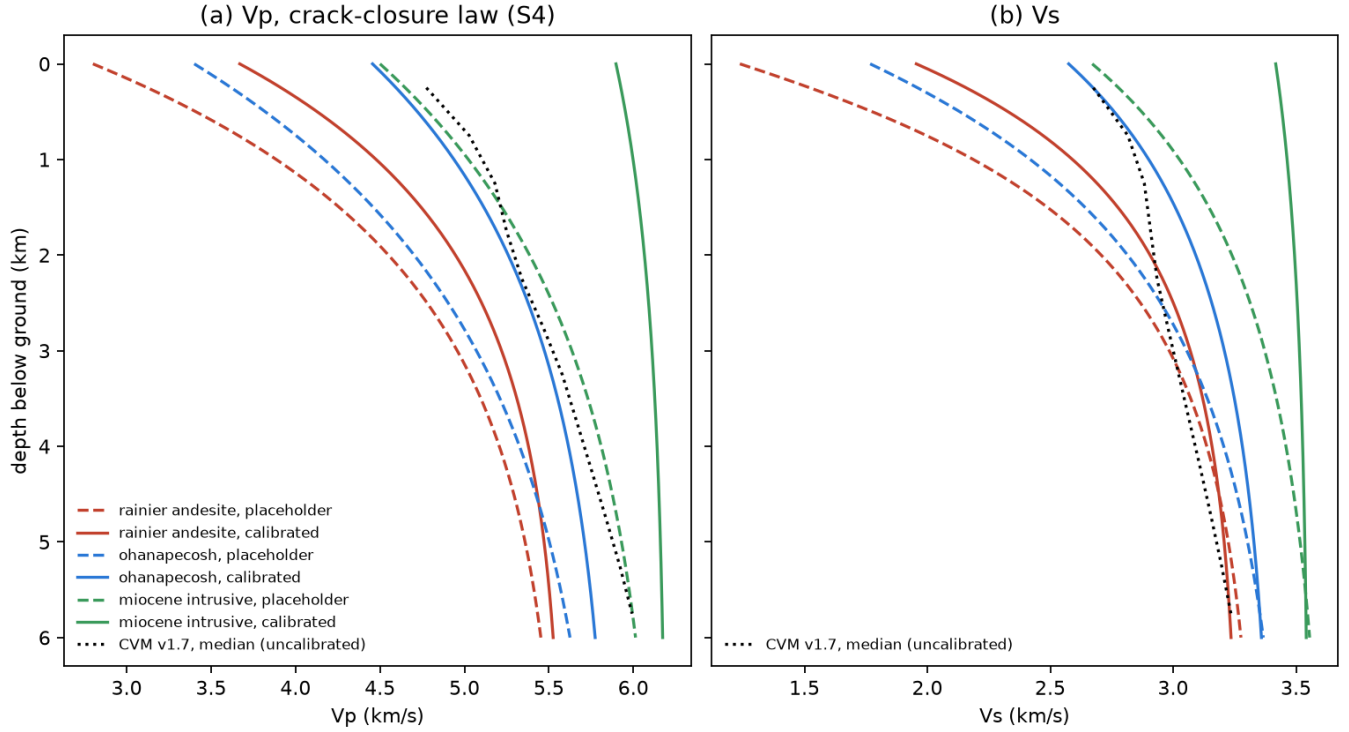


Figure 9. Crack-closure Vp (a) and Vs (b) against depth below ground for three rock units, with table (dashed) and calibrated (solid) parameters. Dotted: median of the Cascadia velocity model over the domain.

Table 13. Residuals after relocating all 88 events in each model (S14, `outputs/relocation/<model>/stats.json`). Depth shifts are relocated minus ComCat, positive deeper. ¹ Fitted on these events; the held-out scores are the fair measure.

Model	P RMS (s)	S RMS (s)	Epicentre shift	
			from ComCat, median (m)	Depth shift, median (m)
PNSN 1D	0.132	0.268	1157	+511
3D, uncalibrated	0.122	0.316	1285	+257
3D, Vs-only depth factor ¹	0.102	0.190	847	+306
3D, depth factors on regional Vp and Vs ¹	0.094	0.190	835	+727
3D, geology multipliers and regional correction (adopted) ¹	0.095	0.190	856	+852

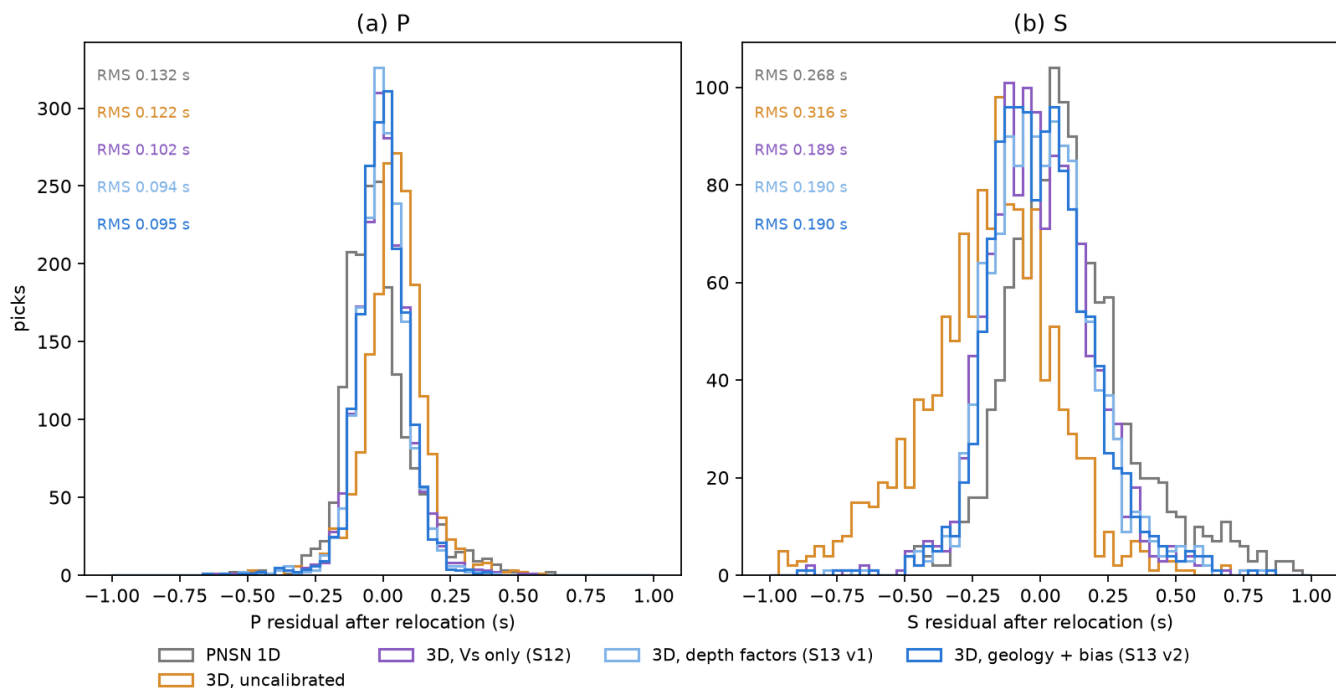


Figure 10. P (a) and S (b) residuals after relocating all events in each model; the RMS of each model is printed in its colour.

- **Beneath Rainier, north of 46.75° N (Table 14).** The iMUSH model finds the Cascadia model's Vs 5–8% too slow at 1–8 km and 3–4% too fast at 11–20 km, the shape of Table 12. With the correction, the difference in Vs is 2% or less below 2 km. Vp/Vs moves from 1.83–1.85 towards the iMUSH value of 1.75–1.77; at 4–8 km our Vp/Vs (1.705) is about 0.05 below it.
- **South of 46.65° N, inside the iMUSH array.** The iMUSH Vs is within 1–2% of the Cascadia model at 1–6 km, and the correction makes our Vs 5–7% too fast. The correction is therefore local to Rainier.
- **Independence.** Both studies use PNSN arrivals from 2015–2016. The iMUSH-only stations and explosions make the comparison largely independent in the south, less so in the north.

Catalogue hypocentres. At the ComCat hypocentres, which were located in a 1D model, the three-dimensional model predicts later arrivals than the 1D model at every station: by 0.30 s on average for P and 0.48 s for S. With each event's mean residual removed, the RMS is 0.132 s (P) and 0.227 s (S) for the 3D model, against 0.159 and 0.287 s for the 1D model. The per-station 3D – 1D delay correlates with the mean 1D residual at 0.67 for both phases (Fig. 12). The 3D structure therefore explains part of what station corrections absorb.

5.7 The fused model

Figs. 13, 14 show the fused model along two sections through the summit, and Fig. 15 compares vertical profiles.

Table 14. North of 46.75° N: mean $\ln(V_{\text{iMUSH}}/V_{\text{model}})$ for Vs and median Vp/Vs (S21, outputs/model_comparison/ulberg2020_by_depth.csv)

Depth below ground (km)	1–2	2–3	3–4	4–6	6–8	11–15	15–20
Vs, regional as distributed	+0.084	+0.073	+0.064	+0.055	+0.046	−0.030	−0.035
Vs, regional with the correction	+0.061	+0.016	−0.010	−0.019	−0.009	−0.005	+0.017
Vp/Vs, iMUSH	1.770	1.759	1.754	1.757	1.755	1.744	1.765
Vp/Vs, regional	1.810	1.828	1.850	1.851	1.851	1.720	1.723
Vp/Vs, rainier3d	1.794	1.767	1.736	1.705	1.705	1.720	1.772

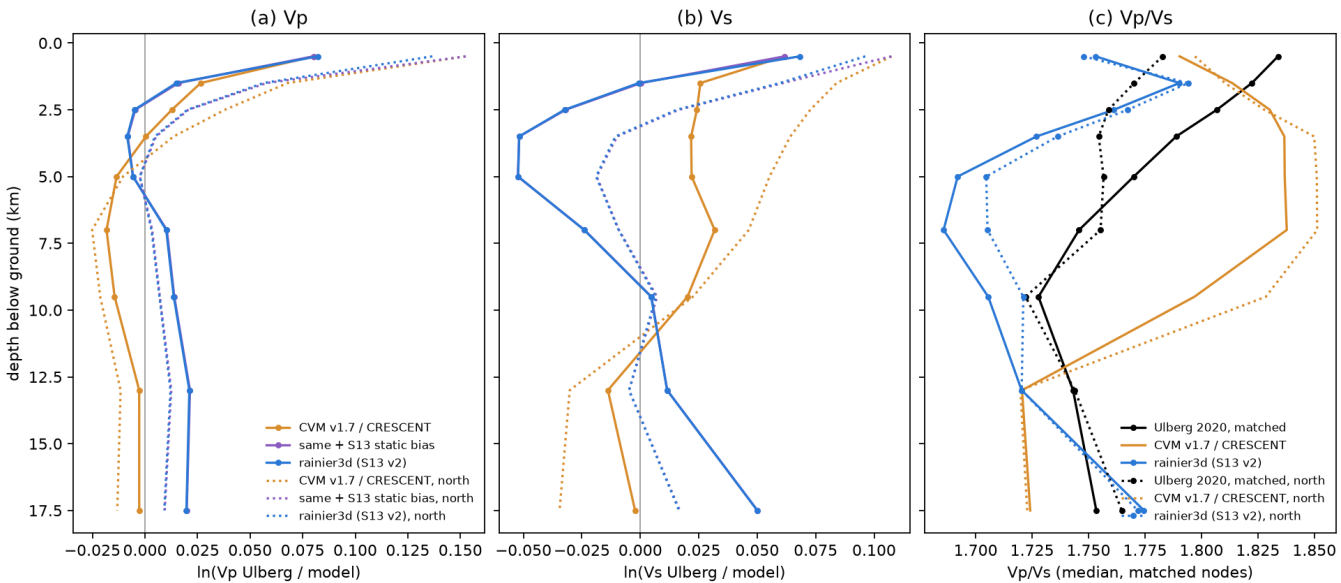


Figure 11. rainier3d and the regional models against the iMUSH tomography. (a, b) Mean $\ln(V_{\text{iMUSH}}/V_{\text{model}})$ against depth below ground, over the box (solid) and north of 46.75° N (dotted). (c) Median Vp/Vs where the iMUSH P and S models are both resolved.

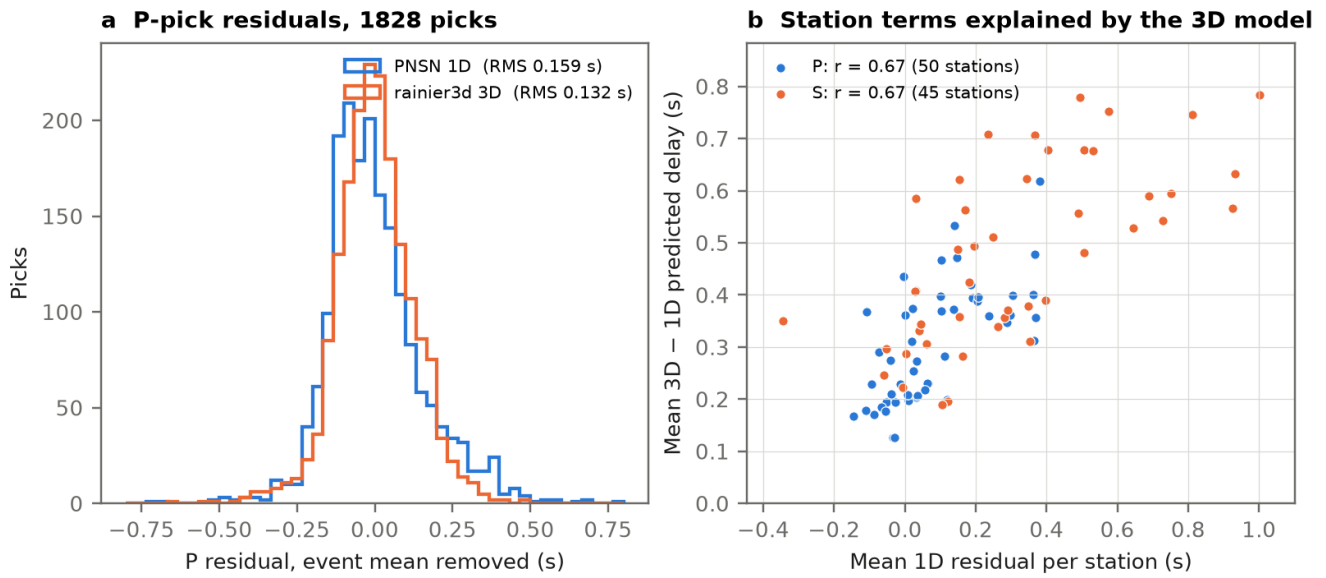


Figure 12. At the ComCat hypocentres: (a) distribution of P residuals after removing each event’s mean, for the PNSN 1D model and the fused model; (b) mean 3D – 1D predicted delay against mean 1D residual for each station with at least five picks.

- **The top kilometre.** L1 is on average 5% faster in Vp and 8% faster in Vs than the regional model. The calibrated rock is stiffer than the shallowest layer of the Cascadia model, a generic near-surface model with a median Vs of 206 m s^{-1} . The travel times constrain this difference only beneath the stations.
- **Plutons.** The Miocene plutons stand out as fast columns to 10 km below sea level.
- **Puget Group.** The Puget Group block west of the summit is slow down to 4 km below sea level.
- **Seismicity.** Summit earthquakes form a column from the edifice to about 3 km below sea level. WRSZ earthquakes concentrate 4–12 km below sea level, 12–18 km west of the summit.
- **Magma body.** The slow body at 7–10 km below sea level is largely removed by the fusion, because neither regional model holds a slow body there at the wavelengths they resolve. Whether a body of the size imaged by Moran et al. (1999) and Pang et al. (2025) belongs in the model is a question for data that resolve it.

6 Geodetic strain and stress at depth

6.1 GNSS data and velocities

Daily GNSS positions come from two archives:

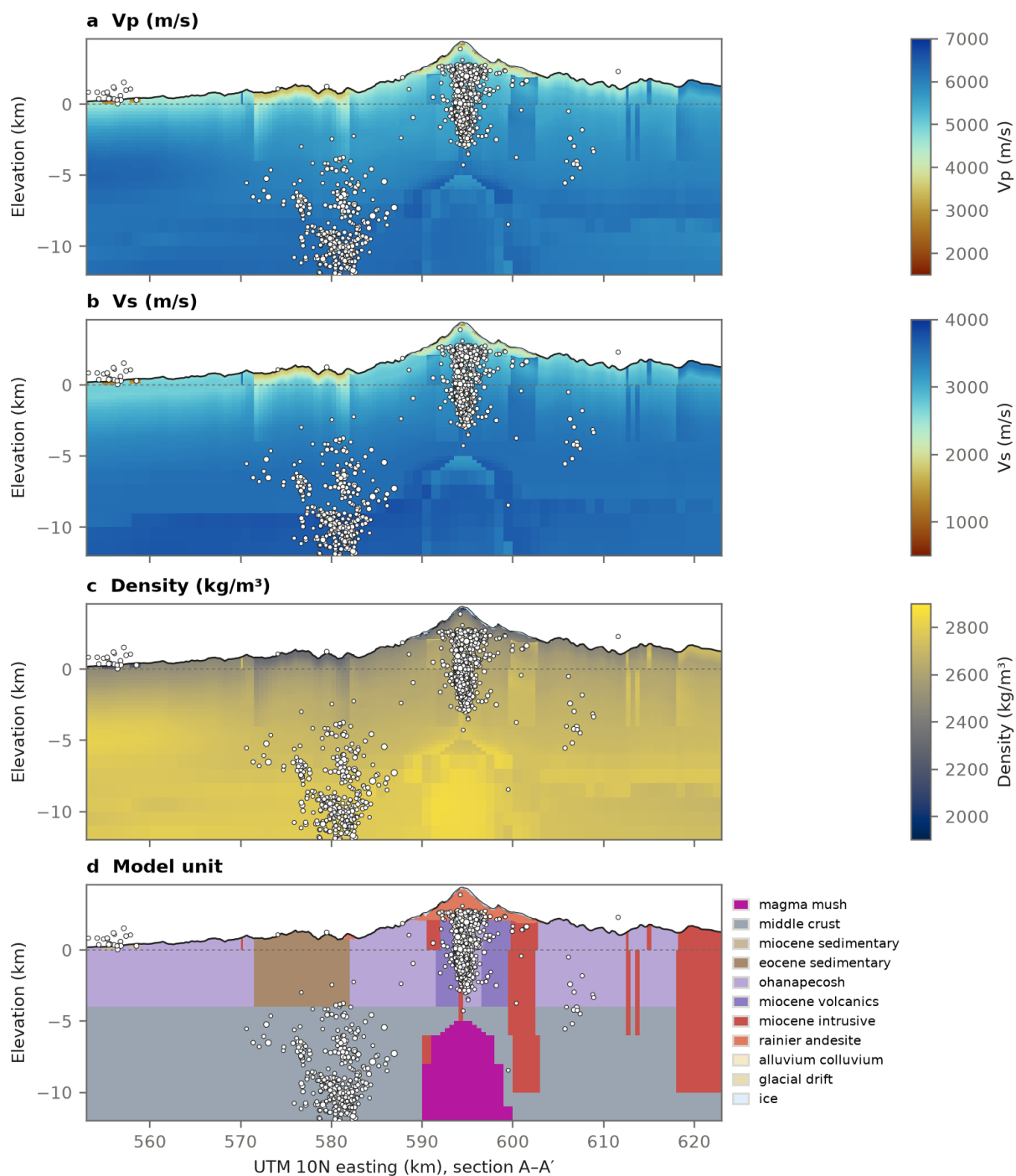


Figure 13. Fused model along A–A' (west–east through the summit): (a) Vp, (b) Vs, (c) density, (d) model units. Light blue at the surface is glacier ice. White dots are PNSN earthquakes within 2 km of the section. The dashed line is sea level.

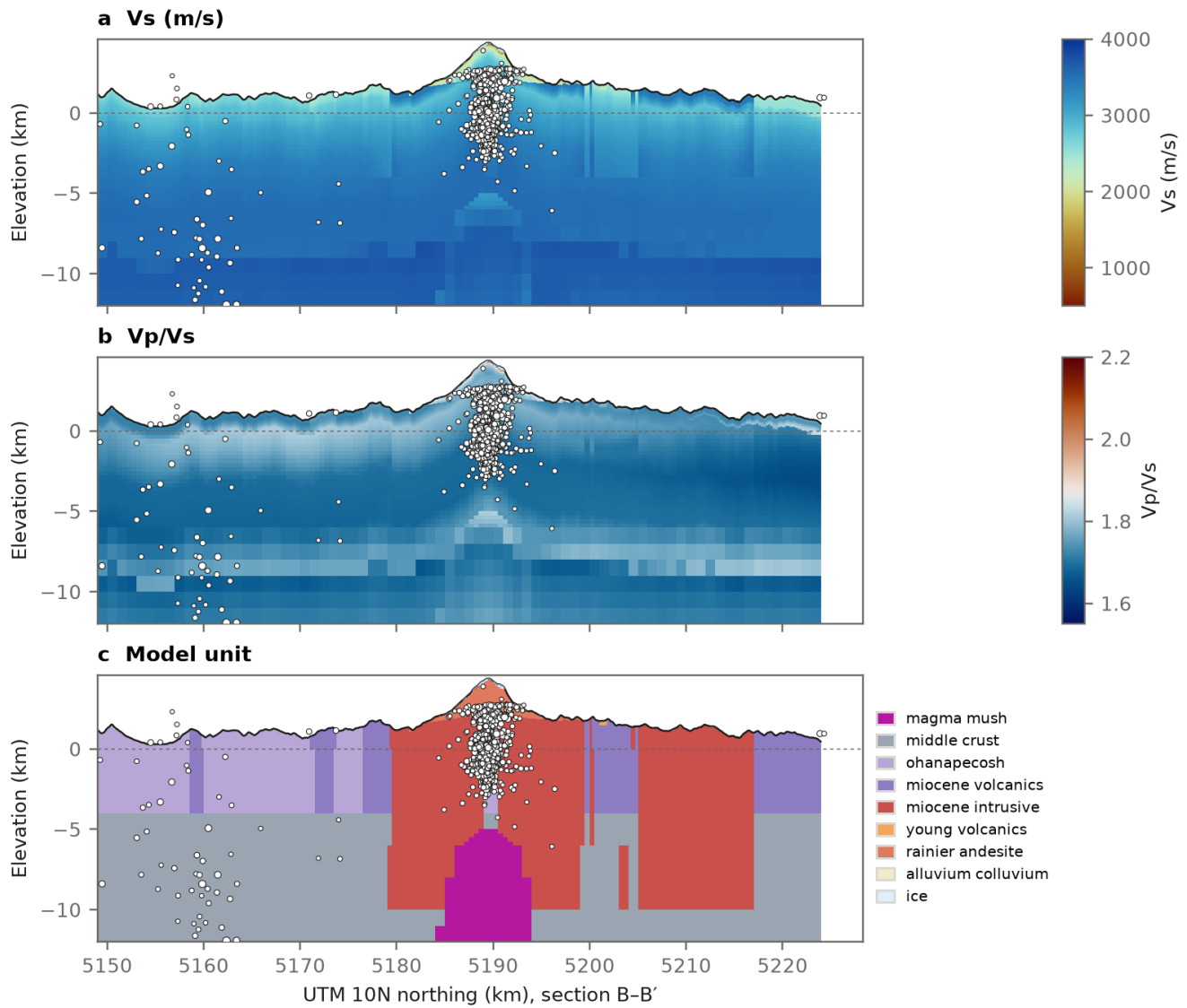


Figure 14. Fused model along B-B' (south-north through the summit): (a) Vs, (b) Vp/Vs, (c) model units. The step in Vp/Vs near 9 km below the ground marks the change from the Cascadia model to CRESCENT with Brocher's Vp.

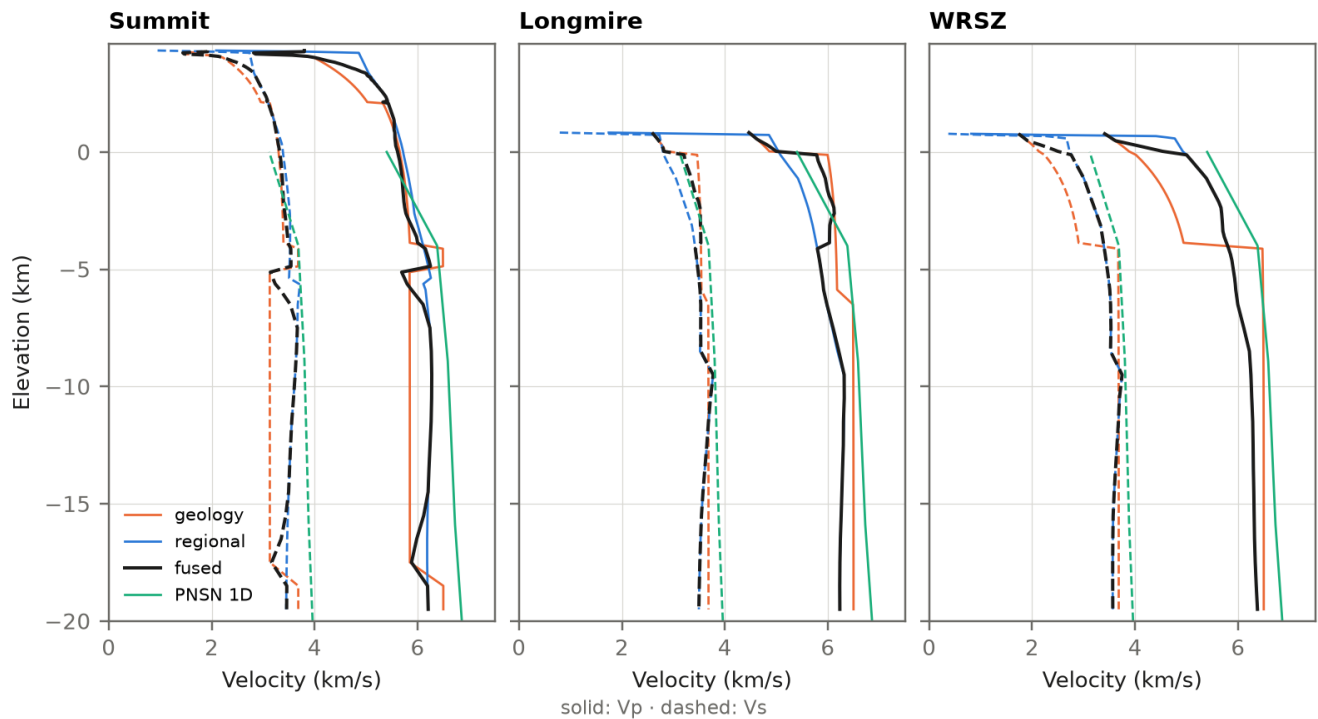


Figure 15. Vertical profiles of Vp (solid) and Vs (dashed) at the summit, at Longmire and in the WRSZ, for the geology, regional and fused models and the PNSN 1D model.

- **PANGA.** The Pacific Northwest Geodetic Array (Central Washington University) is the regional analysis centre (Pacific Northwest Geodetic Array, 2026): 188 sites in a network box of 124.0–119.8° W, 45.6–48.2° N, in its North-America-fixed frame.
- **UNR.** The Nevada Geodetic Laboratory’s IGS20 North-America-fixed series fill in 54 sites that PANGA lacks (Blewitt et al., 2018).

Together they give 1,050,643 site-days in 256 series.

- **Frame alignment.** Twenty sites that both archives process align UNR to PANGA by a translation and rotation fit, with a residual RMS of 0.10 mm yr^{-1} .
- **Velocities.** Station velocities come from MIDAS (Blewitt et al., 2016). Equipment steps are taken from the PANGA fit headers and the UNR steps database.
- **Quality control.** Sites are flagged when their velocity differs by more than 1 mm yr^{-1} from PANGA’s published velocity, or by more than 2.5 mm yr^{-1} from the median of neighbours within 40 km. This flags 22 of 203 sites. They include the high edifice sites MUIR, CSHR, SNRS, PNHG and PNHR; PNHG moves 44 mm yr^{-1} east, which is not tectonic. Flagged sites are kept in the product with their flag and left out of the strain fits.

Table 15. Secular strain rates from uniform-strain fits to MIDAS velocities (S18, `outputs/gnss/summary.json`).

Region	Sites	Dilatation (nanostrain yr ⁻¹)	Maximum shear (nanostrain yr ⁻¹)	Azimuth of $\dot{\epsilon}_1$
WRSZ (polygon + 25 km)	17	-16.6 ± 3.5	11.1	136°
Edifice (within 25 km of the summit, quality-controlled)	6	-27.9 ± 12.4	4.3	171°
Edifice, all sites	11	$+120.5 \pm 11.8$	61.2	103°

The data are fetched again every week, and the products are recomputed (Sect. 9). The numbers below are for the series ending on 1 September 2026, which are frozen in `configs/gnss.yaml`.

6.2 Strain rate

The horizontal strain-rate tensor $\dot{\epsilon}_{ij}$ is estimated from the velocities.

- **Grid.** On a 5 km grid, it comes from a velocity-gradient fit with adaptive Gaussian weights (Shen et al., 2015). The smoothing scale grows until the summed weight and the number of stations reach their minimums.
- **Regions.** For the two regions of interest it is a uniform-strain fit to the stations around them.

The reported components are the dilatation $\dot{\epsilon}_{xx} + \dot{\epsilon}_{yy}$, the maximum shear $\sqrt{(\dot{\epsilon}_{xx} - \dot{\epsilon}_{yy})^2/4 + \dot{\epsilon}_{xy}^2}$, the principal rates and their azimuth, the rotation and the second invariant. The grid also carries the amplitude and peak day of the annual areal strain, the seasonal signal of hydrological loading.

Both regions contract at a few tens of nanostrain per year (Table 15, Fig. 17a), as the rest of the model box does (median -19 nanostrain yr⁻¹). The edifice result depends on quality control: with the flagged summit sites included, it changes sign. On the grid, the summit cell has a dilatation rate of -14.6 nanostrain yr⁻¹ at a smoothing scale of 30 km, because the flagged summit sites are excluded. The annual areal-strain amplitude has a median of 15 nanostrain over the box (5–95%: 4–39), peaking in late August.

6.3 Daily strain and the summit swarms

The daily uniform strain of each region’s stations is fitted to trajectory residuals, with trend, annual and semiannual terms and steps removed per site. Stations joining or leaving the network therefore do not shift the series (Fig. 16). Across the two summit swarms, the 60-day changes are small (Table 16).

Neither swarm shows a transient above about 10 nanostrain in the WRSZ. On the edifice, winter excursions of up to 1500 nanostrain come from the close-spaced high sites. With stations a few kilometres apart, a few millimetres of snow-related antenna error become about 1000 nanostrain of apparent strain. Resolving strain of the edifice or of a magma body needs other instruments: tiltmeters, borehole strainmeters or interferometric synthetic-aperture radar.

Table 16. Median absolute deviation (MAD) of the daily strain and 60-day change in areal strain across the summit swarms, in nanostrain (S18).

Region	Daily scatter (MAD)	13 February 2009 swarm	8 July 2025 swarm
WRSZ	20	+6	+7
Edifice	106	no quality-controlled coverage	−166, inside the 411-nanostrain scatter of the 30-day medians

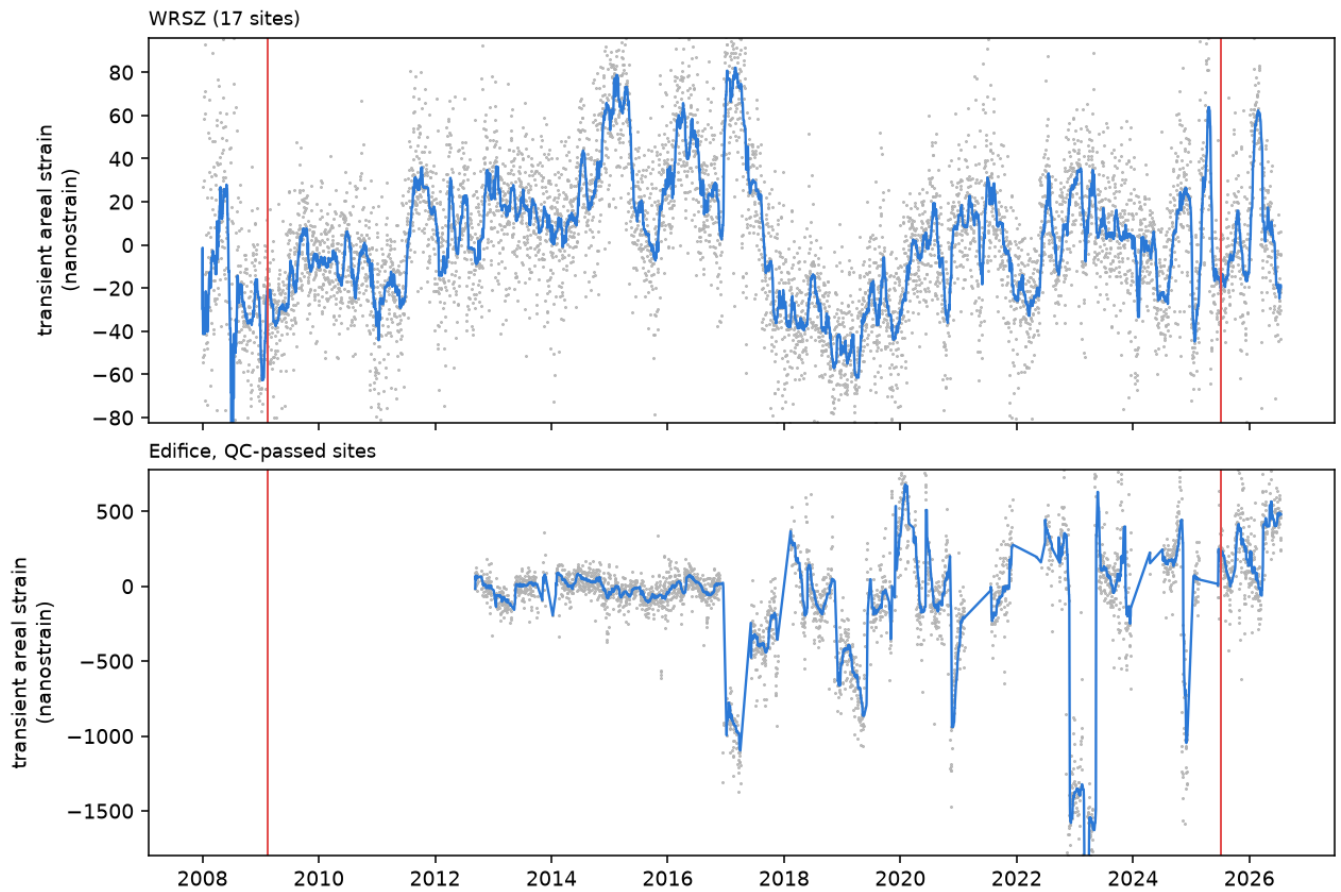


Figure 16. Daily areal strain of the stations around the WRSZ and the edifice, from trajectory residuals (grey), with 30-day medians (blue). Red lines mark the 2009 and 2025 summit swarms.

Table 17. Stress from the edifice load beneath the summit (compression negative; S18, `edifice_load.zarr`).

Elevation (m)	Vertical stress (MPa)	Mean stress (MPa)	Maximum shear (MPa)
0	−38.5	−18.7	15.2
−2000	−24.0	−9.4	11.2
−5000	−13.0	−4.4	6.6
−11,500 (magma-body centre)	−4.9	−1.5	2.6

6.4 Stress from the edifice load

The weight of the edifice is a static load on the crust beneath it.

- **Load.** The edifice is taken as everything above the pre-volcanic surface of the geology model. It is represented by Boussinesq point loads on an elastic half-space (Johnson, 1985): 1131 cells of 500 m, with a total weight of 2.34×10^{15} N (95 km³ of rock at 2500 kg m^{−3}) on a reference plane at 1789 m.
- **Output.** The six stress components, the mean stress, the maximum shear and the principal stresses are computed on the model levels L1–L3.
- **Checks.** The vertical force on a buried plane balances the load within 1%, and the stress field is in equilibrium away from the axis.

The load stress decays from tens of megapascals beneath the summit to a few megapascals at the depth of the magma body (Table 17, Fig. 17b). That is three to four orders of magnitude above the tectonic stress rate implied by the geodetic strain rates (about 1 kPa yr^{−1} for a shear modulus of 30 GPa and 3×10^{-8} yr^{−1}). The load stress therefore sets the orientation of stresses in the upper crust beneath the edifice. The model is a homogeneous half-space with a flat reference plane, and its Poisson’s ratio (0.25) and density (2500 kg m^{−3}) are author choices.

6.5 Strain in the model volume

Script S25 puts two strain fields on the 500 m × 250 m grid of the three-dimensional viewer, from the surface to 20 km below sea level. Both are written with the axes of maximum shortening, so that they can be compared with the fast directions of shear-wave splitting.

Tectonic strain rate at depth. GNSS constrains the horizontal strain rate at the surface only. The horizontal tensor of Table 15 is carried down unchanged through the elastic upper crust. The vertical component follows from plane stress, $\dot{\epsilon}_{zz} = -\nu/(1 - \nu)(\dot{\epsilon}_{xx} + \dot{\epsilon}_{yy})$, with $\nu = 0.25$. Both steps are assumptions, not observations. The tensor is then resolved on vertical planes parallel to the WRSZ. The zone’s strike, N174°E, is the long axis of its 1294 epicentres (elongation 3.1).

At 5 km below sea level inside the WRSZ polygon:

The planes of maximum shear strike N3.5°E and N93.5°E. The WRSZ lies 9° from the first, so the geodetic field loads it almost optimally for right-lateral slip (Table 18, Fig. 18).

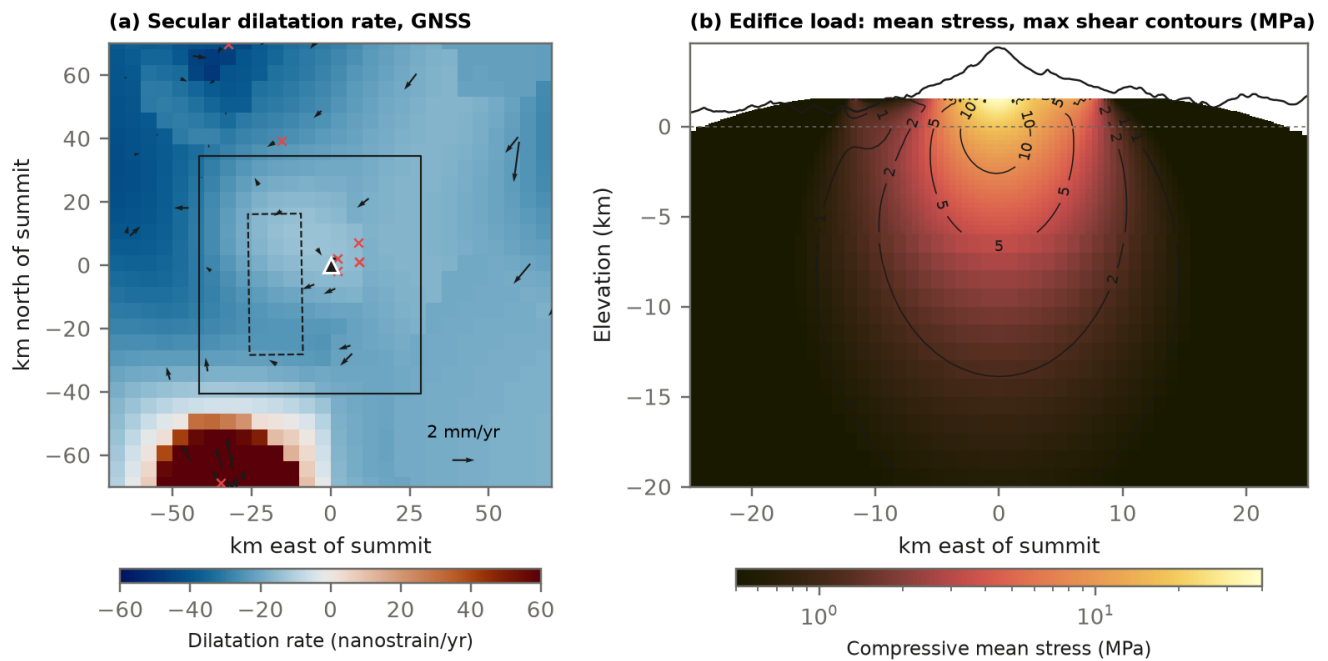


Figure 17. (a) Secular dilatation rate from the GNSS velocities, with the velocities of quality-controlled sites relative to the network median (arrows) and flagged sites (red crosses). The solid box is the model domain and the dashed polygon the WRSZ region. (b) Compressive mean stress from the edifice load along a west–east section through the summit, with contours of maximum shear (MPa).

Table 18. Tectonic strain rate in the WRSZ (S25).

Quantity	Value
Axis of maximum shortening	N48°E
Maximum shear strain rate	10.4 nanostrain yr ⁻¹
Right-lateral shear strain rate on WRSZ-parallel planes	10.0 nanostrain yr ⁻¹
Normal strain rate across the zone	−12.9 nanostrain yr ⁻¹ (contraction)

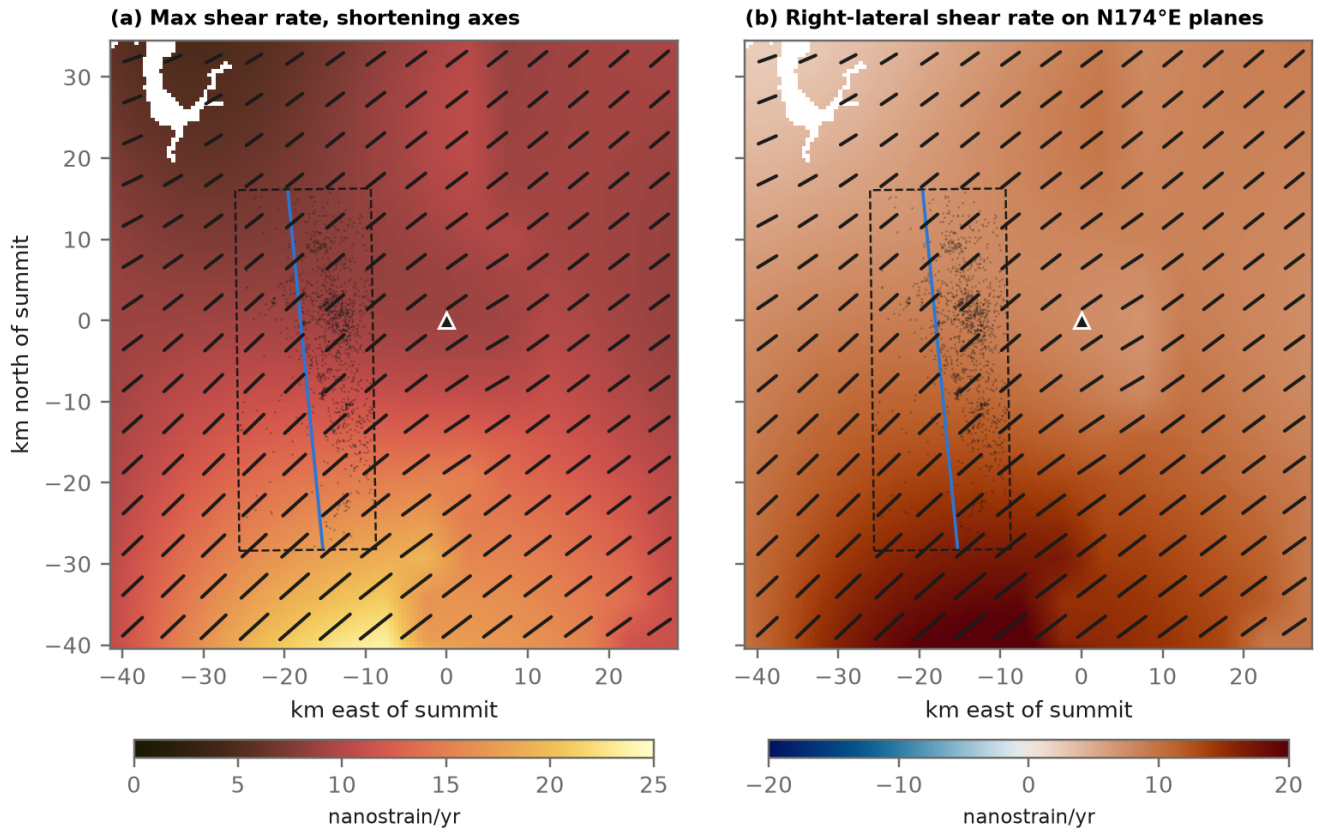


Figure 18. Tectonic strain rate from GNSS, the same at every depth under the stated assumptions: (a) maximum horizontal shear strain rate with the axes of maximum shortening (bars, length scaled with the shear rate); (b) right-lateral shear strain rate on vertical planes parallel to the WRSZ. The dashed polygon is the WRSZ region, grey dots its epicentres, and the blue line the strike fitted to them.

Static strain of the edifice load.

- **Method.** The Boussinesq stress of Sect. 6 is converted to strain with the local stiffness of the velocity model, $\mu = \rho V_S^2$ and $\lambda = \rho V_P^2 - 2\mu$.
- **Cone interior.** Inside the cone above the half-space, that is higher than 1539 m, the stress is taken as the laterally confined overburden, $\sigma_{zz} = -\rho g d$ and $\sigma_h = \nu/(1 - \nu) \sigma_{zz}$. This approximation defines the volumetric strain but no horizontal stress direction.

Beneath the summit (Table 19), the volumetric strain is compressive throughout. It is largest at the base of the cone and decays below. Neither the load nor the geodetic field produces dilatation in the edifice above sea level, where the shallow swarms of the summit occur (Fig. 19). The GNSS areal strain rate there is -15 nanostrain yr^{-1} . Dilatation in that volume would need a source these models do not contain, such as the pressurisation of the hydrothermal system.

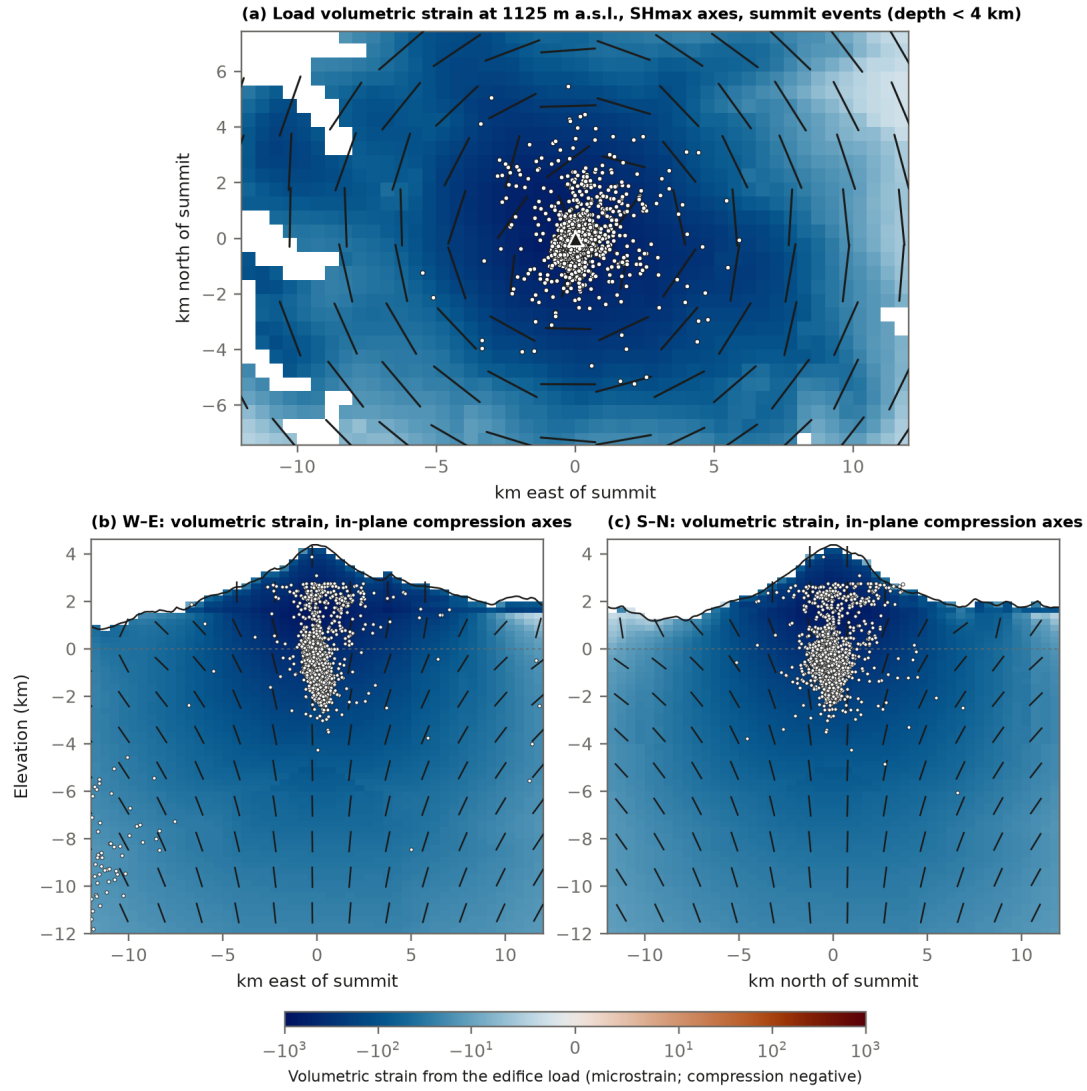


Figure 19. Static strain of the edifice load. (a) Volumetric strain at 1125 m above sea level, with the direction of the most compressive horizontal stress (SHmax, bars) and the summit earthquakes shallower than 4 km (dots). (b, c) West–east and south–north sections through the summit, with the in-plane axis of maximum compression; inside the cone, the confined overburden makes it vertical.

Table 19. Volumetric strain of the edifice load beneath the summit (S25). Above 1539 m the stress is the confined overburden.

Elevation (m)	2500	1500	1000	500	0	−2000	−5000	−11,500
Volumetric strain (mi- crostrain)	−563	−831	−668	−518	−413	−208	−73	−24

SHmax of the load is tangential around the summit at and above sea level and radial from 5 km below sea level down (Fig. 20). The horizontal shear strain of the load reaches 1–30 microstrain within 20 km of the summit, which is 10^3 – 10^4 years of accumulation at the geodetic rates.

The two fields are published on regular grids at fixed elevations (`strain_orientation.csv`):

- the geodetic shortening axes, every 5 km;
- the SHmax of the load, every 2 km within 20 km of the summit.

These grids are for comparison with the fast directions of shear-wave splitting, which aligned cracks orient parallel to the most compressive horizontal stress.

6.5.1 How the strain fields are computed

Both fields are first-order estimates. Neither is computed with Green’s functions of the heterogeneous model or with a rheology other than linear elasticity. This subsection gives the calculation as implemented (`rainier3d.geodesy.load`, `rainier3d.geodesy.volume`, script S25) and then lists what it leaves out.

Tectonic strain rate. The horizontal strain-rate tensor $\dot{\epsilon}_{\alpha\beta}(x, y)$, with $\alpha, \beta \in \{x, y\}$, is the velocity-gradient fit of Sect. 6 at the surface. At depth it is not calculated but assigned:

$$\dot{\epsilon}_{\alpha\beta}(x, y, z) = \dot{\epsilon}_{\alpha\beta}(x, y, 0), \quad \dot{\epsilon}_{zz} = -\frac{\nu}{1-\nu}(\dot{\epsilon}_{xx} + \dot{\epsilon}_{yy}), \quad \dot{\epsilon}_{xz} = \dot{\epsilon}_{yz} = 0. \quad (8)$$

The vertical component follows from a vanishing vertical stress rate (plane stress). This is the thin-plate view of the upper crust: it holds when the sources of deformation are far or deep compared with the depth of interest and the crust is laterally uniform. No source (megathrust locking, fault slip, block rotation) is inverted from the GNSS velocities.

The shear on the WRSZ is resolved as

$$\dot{\gamma} = -\mathbf{s} \cdot \dot{\epsilon} \cdot \mathbf{n}, \quad (9)$$

with $\mathbf{s}$ the horizontal unit vector along the strike and $\mathbf{n}$ the horizontal normal. The sign is positive for right-lateral motion. The strike is the long axis of the covariance of the WRSZ epicentres.

Edifice-load stress. The load is the rock above the pre-volcanic surface of the geology model. Its columns of height h are summed on 500 m cells into vertical point forces $P_k = \rho g h_k A_k$ ($\rho = 2500 \text{ kg m}^{-3}$). The forces act on a flat reference plane at

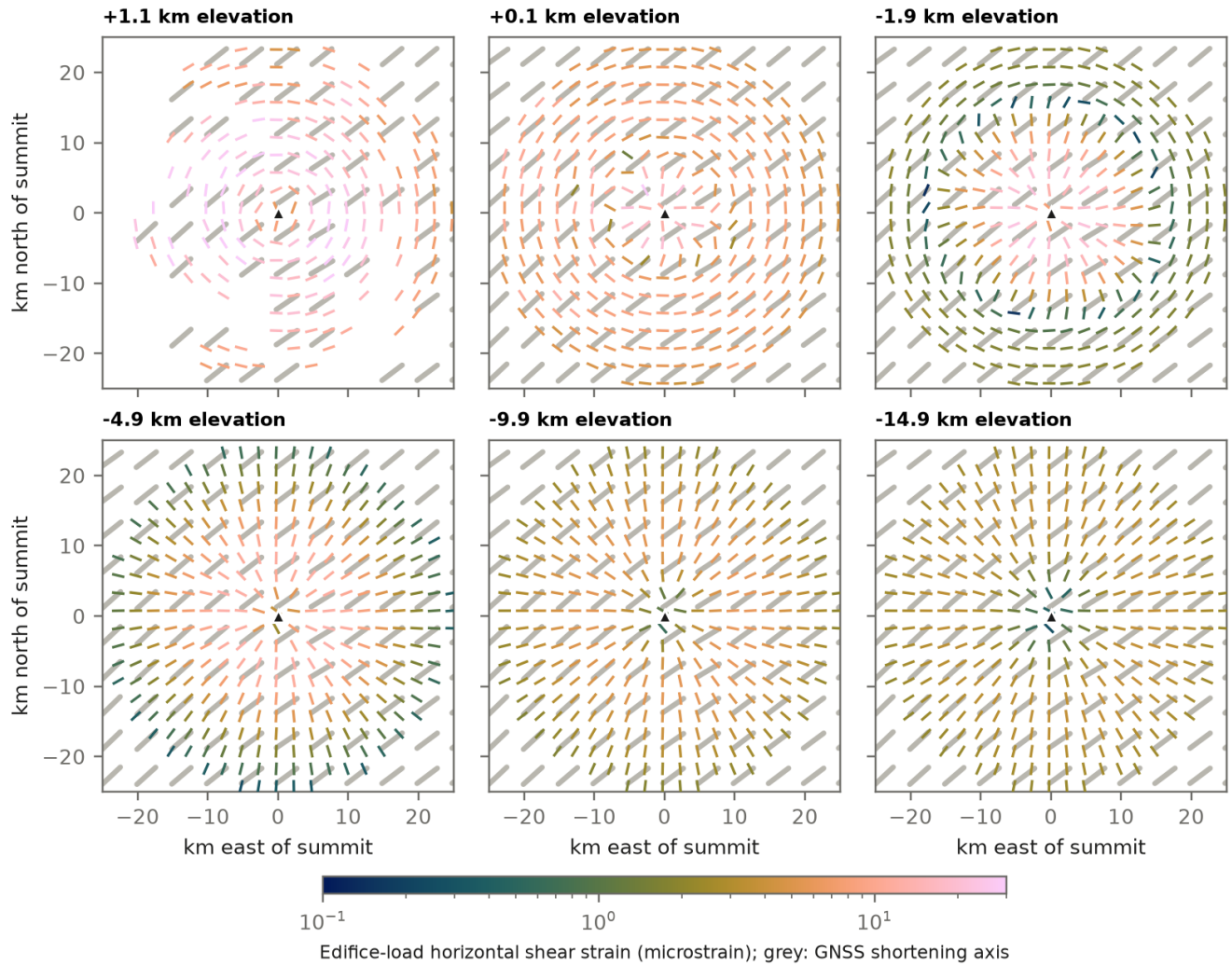


Figure 20. Axes of maximum horizontal shortening at six elevations: GNSS (grey, the same at every depth) and SHmax of the edifice load (coloured by its horizontal shear strain).

485 the mean elevation of the edifice base (1789 m). The stress at a point is the sum over k of the Boussinesq solution for a point force on the surface of a homogeneous, isotropic, linear-elastic half-space (Johnson, 1985). At horizontal distance r and depth z below the plane, with $R = \sqrt{r^2 + z^2}$ and tension positive:

$$\sigma_{zz} = -\frac{3Pz^3}{2\pi R^5}, \quad \sigma_{rz} = -\frac{3Prz^2}{2\pi R^5}, \quad \sigma_{rr} = \frac{P}{2\pi} \left[\frac{1-2\nu}{r^2} \left(1 - \frac{z}{R}\right) - \frac{3r^2z}{R^5} \right], \quad \sigma_{\theta\theta} = -\frac{P(1-2\nu)}{2\pi} \left[\frac{1}{r^2} \left(1 - \frac{z}{R}\right) - \frac{z}{R^3} \right]. \quad (10)$$

These stresses depend on Poisson's ratio ($\nu = 0.25$) but not on the elastic moduli. Points within 250 m below the plane are left out, where the point-force field is singular.

Edifice-load strain. The stress is converted to strain with the isotropic compliance of each cell:

$$\varepsilon_{ij} = \frac{\sigma_{ij}}{2\mu} - \frac{\lambda \sigma_{kk}}{2\mu(3\lambda + 2\mu)} \delta_{ij}, \quad \mu = \rho V_S^2, \quad \lambda = \rho V_P^2 - 2\mu, \quad (11)$$

with V_P , V_S and ρ taken from the fused model.

- **Cone interior.** Above the half-space (higher than 1539 m), the stress is the laterally confined overburden, $\sigma_{zz} = -\rho g d$ and $\sigma_{xx} = \sigma_{yy} = \nu \sigma_{zz} / (1 - \nu)$, with d the depth below the local ground.
- **SHmax.** It is the eigenvector of the horizontal stress tensor $(\sigma_{xx}, \sigma_{yy}, \sigma_{xy})$ with the most compressive eigenvalue. It is undefined in the cone interior, where the horizontal stress is isotropic.

What the calculation does not include.

- **Green's functions of the 3D model.** The stress is that of a uniform half-space, so the stiffness contrasts of the model neither concentrate nor divert it. Applying the local compliance (Eq. 11) makes soft rock strain more under the same stress. The stress field, however, is not in equilibrium in the heterogeneous medium, and its axes do not rotate around stiff plutons or the soft edifice.
- **Topography as part of the elastic body.** The load sits on a flat plane. The cone above it is not part of the elastic domain, and its interior is approximated in one dimension.
- **Rheology beyond linear elasticity.** There is no viscoelastic relaxation of the warm lower crust or of the magma body, no plasticity, no pore-pressure coupling and no thermal stress.
- **A tectonic source model.** The GNSS rates are not explained by locking, fault slip or block motion, so their variation with depth (Eq. 8) is assumed rather than computed.
- **Absolute stress.** The load gives a static stress and GNSS a strain rate. Without the magnitude of the tectonic background stress the two cannot be summed, so their orientations are reported separately.

A consistent model would compute the same quantities in three steps:

1. a static finite-element solution on the model mesh, with gravity acting on the real topography and heterogeneous λ and μ from the fused velocities;

Table 20. Hydrological information in rainier3d.

Component	Layer in rainier3d	Source	Section
Ice	glacier thickness and bed, 5.50 km ³ of ice	IceBoost v2, RGI 6.0	Sect. 4.1
Snow	NDSI at the end of the melt season	Sentinel-2	Sect. 4.2
Surface water	60,174 stream flowlines with Strahler order and mean annual flow	NHDPlus HR	Sect. 4.2
Soil storage	soil thickness (≥ 2 m capped)	SOLUS100	Sect. 4.2
Groundwater	water-table depth, two estimates	Ma et al. (2026); Fan et al. (2017)	Sect. 4.2
Interception and transpiration	canopy height, cover, leaf and plant area, biomass	lidar, Sentinel-2, GEDI	Sect. 4.3
Hydrothermal fluids	alteration intensity of the top ~150 m	1996 helicopter EM survey	Sect. 5.2
Hydrological loading	annual areal-strain amplitude and phase	GNSS	Sect. 6

2. Green’s functions computed in the same 3D elastic model, used to invert the GNSS velocities for their sources before computing strain rates at depth;
3. a viscoelastic lower crust for time-dependent loading.

The load pattern of Fig. 20, tangential beneath the summit and radial at depth, is the qualitative signature of any load on a cone. Heterogeneity would change its magnitudes and rotate its axes near stiffness contrasts. The depth extension of the GNSS axes is the least certain part of the product.

7 Geohydrology

Water controls much of what rainier3d describes:

- the seismic velocities of the shallow crust;
- the conductivity that maps alteration;
- the seasonal loading seen by GNSS;
- the lahars and debris flows of Sect. 8.

The model compiles the hydrological information needed for a hydrological description of the mountain. It holds no hydrological state: it neither models groundwater flow nor assimilates hydrological observations. Table 20 lists what it holds.

The water table. The two estimates differ in level and in pattern.

- **Ma et al. (2026).** The machine-learning estimate of Ma et al. (2026b) keeps the water table within about 20 m of the ground almost everywhere in the box (median 10 m).
- **Fan et al. (2017).** The global groundwater model of Fan et al. (2017) places it hundreds of metres down under ridges.

For the shallow seismic model the choice matters more than either value. A water table separates dry from saturated rock, and saturation raises V_p and V_p/V_s below it. The Ma et al. grid is used under its CC-BY-NC-ND licence for non-commercial research and is not redistributed. Its ensemble uncertainty is served through the HydroGEN platform and needs an account.

535 **Groundwater observations.** No aquifer maps exist for the park. The groundwater study of the upper White River (Fuhrig et al., 2024) is the only park-scale assessment we found. The surficial hydrothermal system described by Frank (1995) is represented only through the alteration it left.

How the layers can enter the seismic model.

- **Fluid substitution.** A water table would separate dry from saturated cells in Gassmann-type fluid substitution.
- 540 – **Valley fill.** Valley fill along the mapped streams would appear as slow, high- V_p/V_s bodies.
- **Hydrothermal core.** A fluid-saturated core would join the alteration field.
- **Loading.** The annual strain signal and the snow and water storage could be compared as a loading model.

None of these couplings is applied in the model.

8 Mass movements: landslides, lahars, debris flows and avalanches

545 Mass movements are Rainier’s most frequent hazard. Debris flows and outburst floods from its glaciers recur on a scale of years (Walder and Driedger, 1995; Legg et al., 2014), rock and ice avalanches fall from its steep upper slopes (Crandell and Fahnestock, 1965), and its lahars have reached the Puget Lowland (Crandell, 1971; Vallance and Scott, 1997). This section defines a catalogue of these events on the model grid: its scope, its record structure and its sources (Table 22). The catalogue adds a time-dependent layer to the model and provides labelled sources for the seismic detection of surface events.

550 **Scope.** Four classes of event:

- landslides, including rock avalanches and edifice collapses;
- lahars;
- debris flows, including glacial outburst floods that bulk into debris flows;
- snow and ice avalanches.

555 For each event, the catalogue records the following:

- class and date, or age for prehistoric deposits;
- source area and runout on the model grid;
- volume where published;
- the triggering conditions where known;
- 560 – the source and its licence.

The catalogue. Script S24 fetches four machine-readable sources, clips them to the box and writes the catalogue to `outputs/mass_movements/` (Table 21, Fig. 21). `docs/mass_movements.md` lists each service call with its endpoint, parameters, paging and cache path. The catalogue has two parts.

- **Flows.** Lahar deposits and debris flows keep their mapped outlines, because their runout is the information. The lahar deposits are the Qv1 units of the 1:100,000 map (Washington Geological Survey, 2026a): the Osceola Mudflow (Vallance and Scott, 1997) covers 79.7 km² of the box, the Electron Mudflow 18.8 km² and other lahar deposits 16.0 km². The debris flows are the flow-type polygons of the Washington State Landslide Inventory Database (Washington Geological Survey, 2026c): 198 from its lidar-protocol mapping and 265 from its compilation of older mapping.
- **Events.** Every other mass movement is a point. The Allstadt et al. compilation of seismogenic mass movements (Allstadt et al., 2017) gives 19 events in the box with a time and a seismic location: the ten 2011 rock and ice avalanches of the Nisqually headwall, five rock falls from Russell Cliff in 1989 and 1992, a 2010 snow avalanche, a 2014 ice avalanche, and the August 2015 rock fall and debris flow. The inventory gives 1,624 landslide polygons of other types and seven recent-landslide points. A polygon becomes a point at its crown, the highest point of its outline, where failure began. The crown is sampled every 2 m on a 1 m window of the USGS 3DEP elevation service around each polygon (U.S. Geological Survey, 2026c), which returns the best 3DEP source at each point: 994 crowns lie on 1 m lidar, 441 on 3 m and 186 on 10 m data. Three windows were refused by the service on every attempt; those crowns are placed on the 30 m DEM and flagged. The compilation repeats three protocol deposits; those copies are dropped.

The inventory records a failure depth for every lidar-protocol deposit (453 deposits in the box, median 14.6 m), so these deposits could later be given a volume in the model. The 1998 lahar inundation zones (Hoblitt et al., 1998; Schilling et al., 2008) are kept as a separate hazard layer, not as events: the case 1 zone covers 964 km² of the box. The Exotic Seismic Events Catalog version 3 (Collins et al., 2025) extends the seismic compilation to 2025 but has no scripted download yet.

Other machine-readable sources. The U.S. Landslide Inventory (Belair et al., 2025; Mirus et al., 2020) compiles the same state data nationally, and the compilation layer includes the park-wide mapping of 448 mass movements over 37 km² by Riedel and Dorsch (2016). PNSN classifies surface events in its own catalogue, but these classes are not in ComCat: within 25 km of the summit, ComCat lists 14,288 events and none of them is a surface event. The curated PNSN dataset of Ni et al. (2023) and the classifiers of Kharita et al. (2026) give labelled waveforms for training detectors.

Sources in the literature. Other sources are reports and papers, from which events are digitised.

- **Lahars.** Holocene lahars and their deposits: Crandell (1971), Scott et al. (1995), and the Osceola Mudflow (Vallance and Scott, 1997).
- **Rock avalanches.** The 1963 rock avalanches from Little Tahoma Peak (Crandell and Fahnestock, 1965).
- **Debris flows and outburst floods.** Along Tahoma Creek (Walder and Driedger, 1994, 1995), and debris-flow initiation in proglacial gullies (Legg et al., 2014).
- **River response.** River aggradation measured by repeat lidar (Anderson and Pitlick, 2014).

Table 21. The mass-movement catalogue in the model box (S24, `outputs/mass_movements/summary.csv`, run of 2026-09-25). Seismic events are from Allstadt et al. (2017); mapped and recent events from Washington Geological Survey (2026c). Of the 1,650 events, 377 are dated; 351 of these are landslides of the January 2009 storm in the compilation.

Part	Class	Source	Count	Area (km ²)
Flows	Osceola Mudflow	Washington Geological Survey (2026a)	1	79.7
	Electron Mudflow	Washington Geological Survey (2026a)	1	18.8
	Other lahar deposits	Washington Geological Survey (2026a)	1	16.0
	Debris flows	Washington Geological Survey (2026c)	463	11.2
Events	Rock fall, rock and ice avalanche	seismic 16; mapped 40	56	
	Snow or ice avalanche	seismic	2	
	Debris flow, outburst flood	seismic	1	
	Slide, debris slide	mapped 317; recent 7	324	
	Complex or unknown type	mapped	1,267	

Table 22. Sources for the mass-movement catalogue.

Class	Main sources	Machine-readable
Landslides, rock avalanches	WGS inventory (with NPS mapping); U.S. Landslide Inventory; ESEC	yes (polygons; seismic events)
Lahars	Crandell (1971); Scott et al. (1995); Vallance and Scott (1997); USGS hazard zones	zones only; events from the literature
Debris flows, outburst floods	Walder and Driedger (1994, 1995); Legg et al. (2014); ESEC; PNSN surface events	partly
Snow and ice avalanches	Northwest Avalanche Center; PNSN surface events	on request

- **Seismic signatures.** Seismic and acoustic signatures of mass movements at volcanoes (Allstadt et al., 2018), and the lahar-detection network of the Cascades Volcano Observatory (Kramer et al., 2024).
- **Snow avalanches.** Observations are held by the Northwest Avalanche Center (Northwest Avalanche Center, 2026), with programmatic access on request.

Each record carries its source, like every other layer. Records from the literature are to be entered from their tables with a citation to the page.

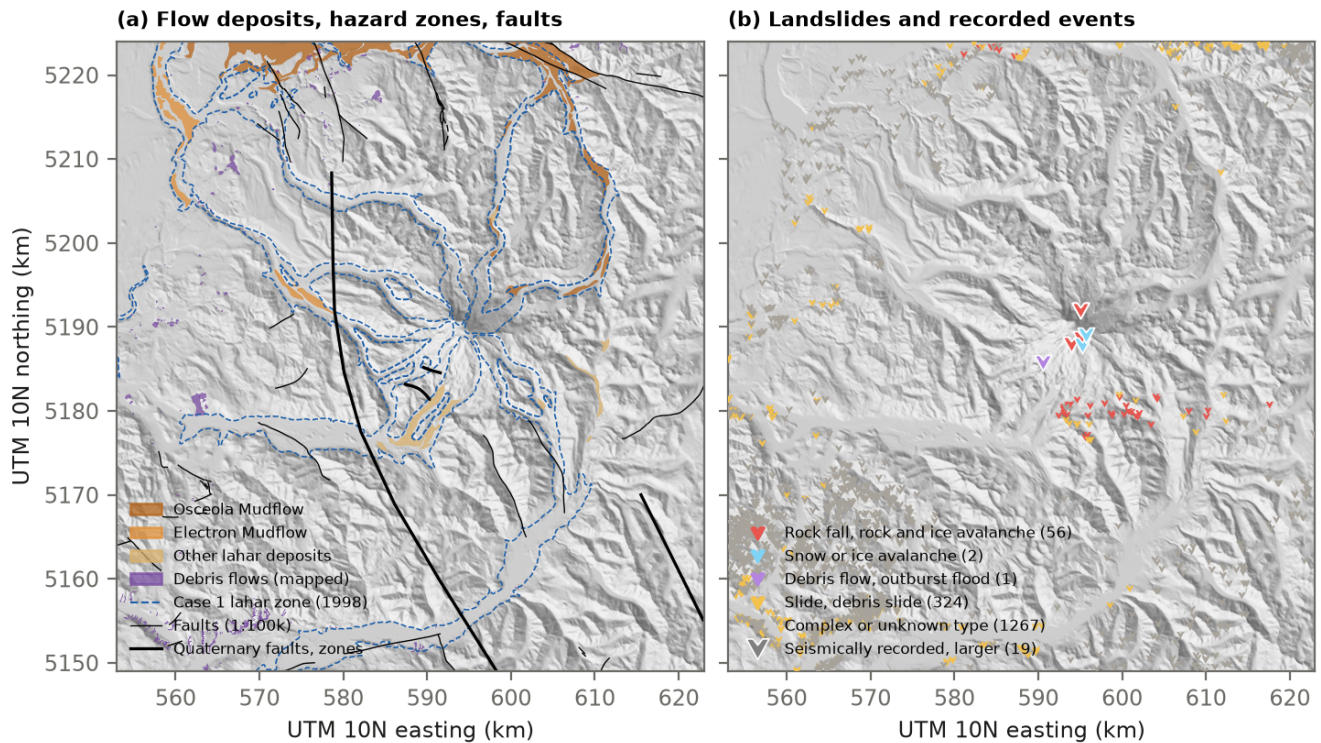


Figure 21. Mass movements and faults. (a) Lahar deposits and mapped debris flows, the case 1 lahar inundation zone of 1998 (dashed), and faults of the 1:100,000 map (thin) and of the Quaternary fault layer (thick). (b) Events by class, drawn as downward chevrons as in the viewer: landslides at their crown (small) and seismically recorded events (large, white rim).

600 9 Accessing the model

9.1 Products and the client

The derived products are published as assets of the release `products-v1.0.0` of the code repository, under CC-BY 4.0 (Table 23). The archive names and their SHA-256 checksums are listed in the release file `SHA256SUMS` and in the package catalogue `src/rainier3d/products.json`.

605 The archives need no software from this project. The subsurface model is downloaded, checked and opened with standard tools:

```
B=https://github.com/Denolle-Lab/mt-rainier-digital-model/releases/download/products-v1.0.0
curl -LO $B/SHA256SUMS -LO $B/rainier3d_model.zarr.zip -LO $B/rainier3d_grids.zip
sha256sum -c SHA256SUMS --ignore-missing # or: shasum -a 256 -c (macOS)
unzip rainier3d_model.zarr.zip && unzip rainier3d_grids.zip
```

Table 23. Downloadable products.

Product	Size	Content
model	110 MB	the subsurface model, <code>model.zarr</code> : levels L1–L3 (Table 2) with Vp, Vs, density, Qp, Qs, unit, alteration, and the geology and regional inputs, plus the surface node; variables from sources that forbid redistribution are removed
grids	55 MB	the subsurface model on one uniform 500 m grid: CF netCDF, EMC-style netCDF, NonLinLoc P and S grids
strain_3d	72 MB	strain in the volume (Sect. 6.5): GNSS strain rate carried down and edifice-load strain, with their orientations
edifice_load	115 MB	stress from the edifice load on L1–L3
alteration	1 MB	alteration from the helicopter electromagnetic survey (Sect. 5.2)
mass_movements	2 MB	the mass-movement catalogue (Sect. 8), GeoPackage and CSV
gnss	4 MB	station velocities with quality flags, strain-rate grid, daily regional strain series and the download manifest; refreshed weekly in the rolling release gnss-latest, with dated copies

```
import xarray as xr
tree = xr.open_datatree("model.zarr", engine="zarr", consolidated=False)
vs = tree["L2"].to_dataset().vs.interp(x=594500, y=5189500, z=-2000) # m/s, 2 km below sea level
g = xr.open_dataset("grids/rainier3d_fused_500m.nc") # whole model, 500 m spacing
```

In `model.zarr`, cells above the ground are empty (NaN), and the unit codes are those of `configs/units.yaml`.

A client is installed with the package. It is available as the `rainier3d` command and as the Python module `rainier3d.api`. It downloads a product once, checks its checksum, caches it, and writes grids for other codes:

```
pip install "git+https://github.com/Denolle-Lab/mt-rainier-digital-model"
rainier3d list # products, versions and licences
rainier3d fetch model # prints the folder that holds model.zarr
rainier3d export model --format specfem --bbox -122.0 46.7 -121.6 47.0 \
    --dx 250 --dz 250 --zmin -10000 --out out/tomography_model.xyz
rainier3d export model --format nll --dx 500 --dz 500 --out out/nll/rainier3d
rainier3d export surface --layers elevation soil_thickness --out out/surface/ # GeoTIFFs
rainier3d sample --lon -121.76 --lat 46.85 --depth 5000 # values at one point
rainier3d fetch gnss # latest weekly strain product

import rainier3d.api as r3
tree = r3.open_model() # downloads once, then reads the cache
g = r3.grid(tree, bbox=(-122.0, 46.7, -121.6, 47.0), dx=250, dz=250, zmin=-10000)
r3.export(g, "netcdf", "out/rainier3d_250m.nc") # also nll, emc, specfem, pylith, csv
```

610 The repository file `docs/products.md` gives examples for every product.

Table 24. Export formats of `rainier3d` export.

Format	Coordinates	Use
CF netCDF4	UTM 10N x, y (m); z = elevation (m, up)	eikonal solvers (pykonal, fteikpy), custom codes
NonLinLoc 3D grids (P, S), slowness $\times$ cell size	UTM 10N km, depth down below sea level, <code>TRANSFORM</code> NONE	Grid2Time, NLLoc
EMC netCDF3	longitude, latitude, depth below sea level (km)	EarthScope Earth Model Collaboration tools
SPECFEM3D <code>tomography_model.xyz</code>	UTM 10N m, elevation up, x fastest	SPECFEM3D Cartesian
PyLith spatial database (SimpleGridDB): density, Vs, Vp, with a parameter snippet	UTM 10N m, elevation up	PyLith static and quasi-static elasticity
CSV	x, y, z and variables, one row per node	spreadsheets, GIS
GeoTIFF (surface layers)	UTM 10N, 100 m	GIS

Resampled grids are interpolated linearly within each level. Cells above the ground take the value of the first rock cell beneath them, so receivers at the surface sit in rock, and a variable `air` flags them. Table 24 lists the formats.

9.2 Locating earthquakes in the three-dimensional model

Of the 15,660 PNSN earthquakes since 1980 in the box, 500 have ComCat hypocentres at or above the ground surface, mostly shallow edifice events. Those events were located in one-dimensional models that know neither the topography nor the slow edifice. Script S26 relocates the catalogue with NonLinLoc (Lomax et al., 2000) twice: once in the PNSN 1D model and once in `rainier3d`. Both runs use the same picks, stations, grid geometry and settings, so the difference between the two catalogues isolates the velocity model. ComCat is kept as a third reference. The settings are:

- **Grids.** Equal spacing in all directions (500 m here; 250 m is practical near the edifice). The 1D model is written on the same grid as the 3D model.
- **Keep hypocentres in rock.** The search volume is masked below the ground with `LOCTOPO_SURFACE`, reading an ASCII GMT grid of the DEM in kilometres on the UTM frame. With `TRANS NONE`, NonLinLoc compares $-z$ with that grid. The exported velocities are left unchanged, and air cells carry the rock velocity below them, so travel times are unbiased. Slow air above a rock skin is an alternative, but the skin must be at least two cells thick: with a one-cell 500 m skin, Grid2Time’s finite-difference start box reaches the slow air around summit stations and delays P times at station OBSR by 0.41 s on average.

Table 25. Relocation of the 2023–2025 PNSN catalogue, magnitude ≥ 1 (S26, outputs/catalog/summary.json and catalog_relocated.csv). Rows below “Located” are for the 340 grade A and B events located in both models. Summit events lie within 3 km of the summit and WRSZ events 8–25 km west of it, both by ComCat epicentre.

	ComCat	NonLinLoc, PNSN 1D	NonLinLoc, rainier3d
Located (of 370)	–	347	353
Rejected: base of the volume / side	–	3 / 20	12 / 5
Median RMS, grade A and B (s)	–	0.119	0.109
Median depth sd; largest horizontal uncertainty (km)	–	0.61; 0.61	0.68; 0.57
Above the ground	1	4, pinned at the mask	0
Summit (165 events): median elevation; 5–95% range (km)	–0.31; –1.29 to 2.18	–0.28; –0.90 to 0.77	–0.91; –1.82 to 0.65
WRSZ (129 events): median elevation; 5–95% range (km)	–8.8; –14.7 to –3.7	–9.6; –16.0 to –3.4	–9.5; –15.2 to –4.5

- **Stations.** In the UTM kilometre frame, with depth = $-\text{elevation}/1000$ (GTSRCE <sta> XYZ <x_km> <y_km> <–elev_km> 0.0).
- **Travel times and location.** GTMODE GRID3D ANGLES_NO; the equal-differential-time likelihood and oct-tree search. Gaussian pick errors are 0.14 s for P and 0.23 s for S, as in the calibration, and model errors grow with travel time (LOGGAU2 0.02 0.05 0.5).
- **Quality.** Grade A: gap $< 180^\circ$, at least eight phases and depth standard deviation < 2 km. Grade B: gap $< 250^\circ$ and at least six phases. Grade C: the rest.

The 2023–2025 catalogue. We applied S26 to the 371 PNSN earthquakes of magnitude 1 or larger in the box from 2023 to 2025. They carry 20,425 analyst picks (12,546 P and 7,879 S) at 45 stations inside the box, and 370 have at least six picks. NonLinLoc locates 347 events in the 1D model and 353 in rainier3d. The others are rejected because their most likely location lies on the edge of the search volume (Table 25):

- in rainier3d, 12 rejected events lie at its base, at 18.7 km below sea level, the deepest level searched;
- the remaining rejections, in either model, are epicentres at the edge of the box, beyond which there are few stations.

What changes. The residuals fall by 8% in rainier3d (median RMS 0.109 against 0.119 s). The events move a median 0.6 km horizontally and 0.44 km deeper than in the 1D model (10th–90th percentile -0.55 to $+1.46$ km).

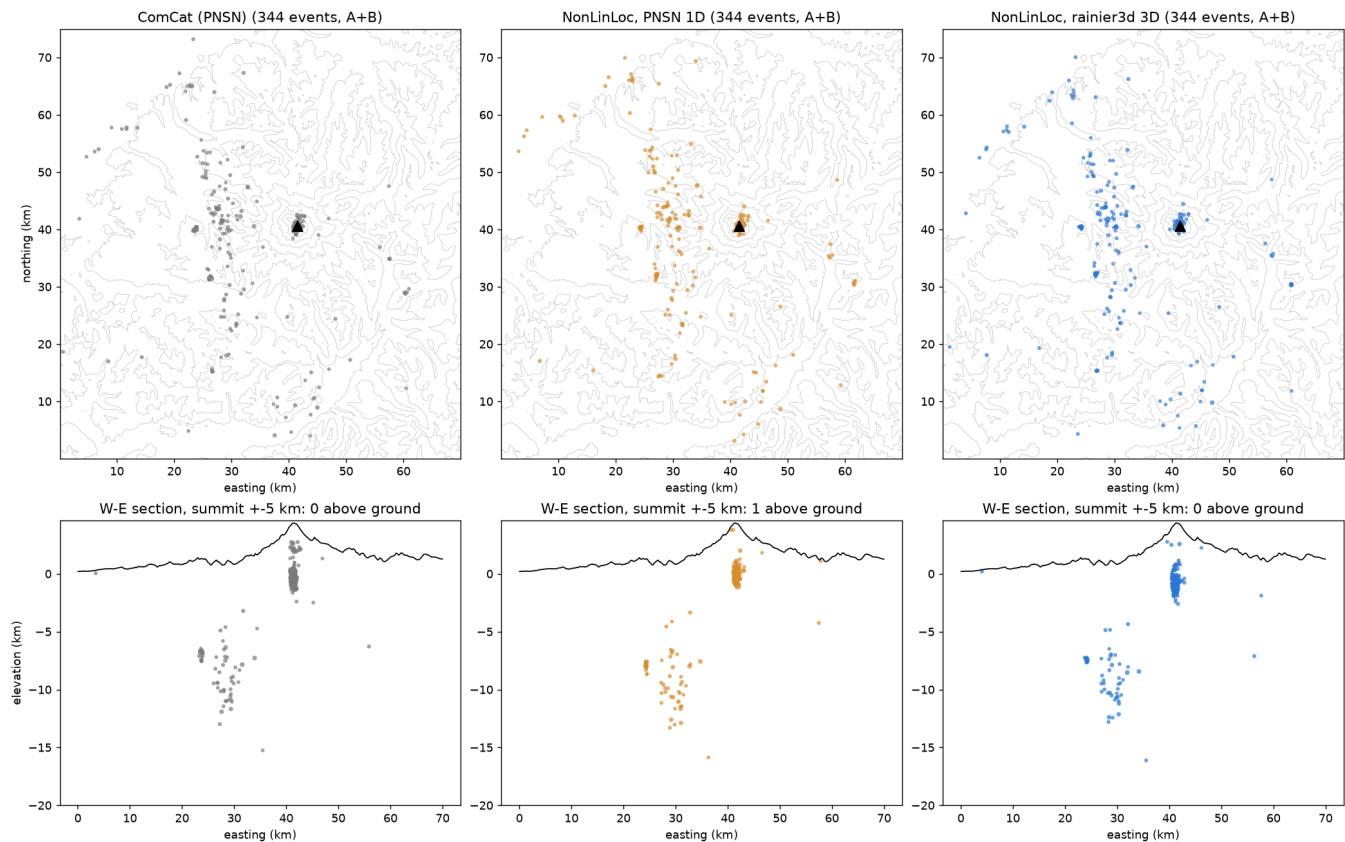


Figure 22. The 2023–2025 PNSN catalogue (magnitude ≥ 1 ; the 344 events graded A or B in rainier3d) as located by ComCat, by NonLinLoc in the PNSN 1D model and by NonLinLoc in rainier3d. Top: epicentres over elevation contours. Bottom: west–east sections within 5 km of the summit, with the ground profile, and the number of events above the ground.

- **Summit.** The largest change is under the summit. The 1D model gathers the events near sea level, over 1.7 km (5–95%). rainier3d places them 0.76 km deeper (median) and spreads them over 2.5 km. In the 1D model, four events rise until they meet the topography mask: ComCat places them 2.8–7.1 km below sea level, and rainier3d 1.2–9.9 km below it.
- **WRSZ.** The median depth changes little (-0.16 km), and the depth range narrows from 12.6 to 10.7 km.
- **Edges.** Epicentres near the northwestern and southeastern edges of the box shift outward in both models, where the station coverage ends.

In this period ComCat places one event above the ground. The problem of the 500 above-ground events therefore lies in the older part of the catalogue, which S26 can relocate once its picks are cached (Sect. 10).

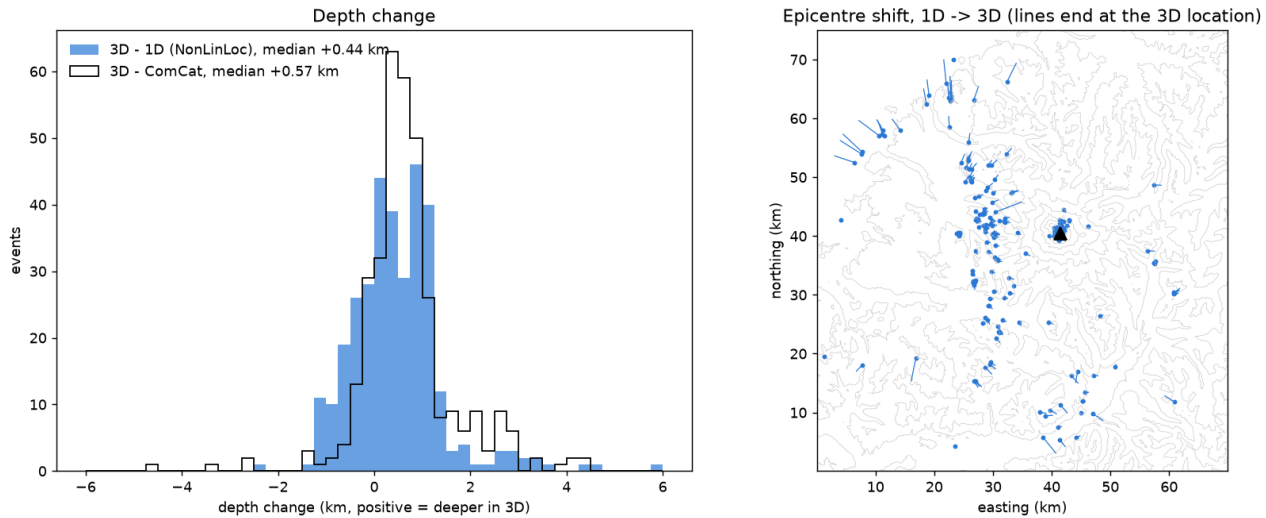


Figure 23. (a) Depth change of each event between rainier3d and the 1D model (filled) and between rainier3d and ComCat (outline). (b) Epicentre shifts from the 1D to the 3D location.

9.3 The three-dimensional viewer

The viewer runs in a web browser, including on phones. It is a React and three.js application (MIT licence) published at <https://denolle-lab.github.io/mt-rainier-digital-model/>, and it draws the following on the terrain:

- the surface layers of Sect. 4;
- the alteration field and the apparent magnetisation;
- the seismic, geodetic, infrasound, tiltmeter and fibre sensors, with permanent and temporary networks separated;
- the PNSN seismicity;
- the mass movements of Sect. 8: the flow deposits as a draped layer and the events as points on the ground, filtered by class and date from the legend.

Below the ground it shows V_s , V_p , V_p/V_s , density, units, alteration and the strain fields of Sect. 6.5, on a vertical section along the terrain cut and on a horizontal depth slice. For the strain fields, the depth slice also carries their orientation bars. A panel under the subsurface controls shows the relocated catalogue of Sect. 9.2, as located by ComCat, in the 1D model and in rainier3d, with optional lines from each 1D location to its 3D location. Its map data are built by scripts S8, S11, S24, S25 and S26 and published as a release asset named in `web/viewer/DATA_RELEASE`.

10 Limitations

- **Rock-physics parameters.** The unit parameters of Table 8 are chosen within published ranges and scaled by three global multipliers, of which only V_0 is resolved by the travel times. The alteration factors and the Q rule are likewise

author choices. The planned replacement is a table of laboratory and field measurements on Cascade and analogue volcanic rocks (Watters et al., 2000; Heap and Violay, 2021), against which the calibrated V_0 can be tested. Per-lithology multipliers need more stations on plutonic and sedimentary units.

- 670 – **Geometry by rules.** Contacts are vertical and unit bases are flat. Structural modelling from the map contacts and orientation measurements would replace the column rules.
- **Hidden alteration and deep bodies.**
 - The alteration field covers only the top ~150 m. The buried alteration of the upper west flank (Finn et al., 2001) is not represented.
 - 675 – The magma body is a prescribed ellipsoid that the fusion largely removes.
 - The Southern Washington Cascades Conductor (Stanley et al., 1996) is not represented.
- **Regional correction.** It is one depth profile for the whole box. The iMUSH comparison shows that it holds beneath Rainier but makes Vs 5–7% too fast in the south, so it should vary laterally.
- **Top 300 m.** The calibrated rock is 21% faster in V_p and 30% faster in Vs than the shallowest layer of the Cascadia model, and the travel times constrain it only beneath the stations. Near-surface Vs from dense nodal and fibre arrays, or station terms estimated with a prior on their size, can settle which is right. Station terms are not estimated in the calibration.
- 680 – **Depth coverage of the calibration.** Only 30 of the 88 events are deeper than 11 km. Larger pick sets, such as the curated PNSN dataset (Ni et al., 2023), and Rainier-specific models (Obrebski et al., 2015; Flinders and Shen, 2017) would constrain the deep correction.
- 685 – **Regional model below 9.9 km.** The regional model there is CRESCENT Vs with Brocher’s V_p ; the deep level of the Cascadia model (10.8–59.4 km) is not used.
- **Resolution.** L1 is 250 m $\times$ 50 m, so thin deposits fall below the cell size.
- **Glaciers.** IceBoost exceeds the 1981 radar thicknesses on Emmons and Winthrop glaciers; its total should be compared with the lidar-based ice volume of Sisson et al. (2011).
- 690 – **Geodesy and strain.** The GNSS network does not resolve strain on the edifice. The strain in the volume uses no Green’s functions of the heterogeneous model and no rheology beyond linear elasticity (Sect. 6.5.1):
 - the geodetic strain rate at depth is assigned (depth-invariant horizontal rate, plane stress), not inverted from sources;
 - the edifice-load stress is that of a uniform half-space with a flat surface, converted to strain with the local stiffness, with a confined-overburden approximation inside the cone.
- 695 – **Hydrology.** The model has no hydrological state (Sect. 7).

11 Conclusions

rainier3d assembles, in one reproducible structure, what is openly known about the subsurface and surface of Mount Rainier.

- **Subsurface.** A geology-driven velocity model, fused with the regional tomography, calibrated on PNSN P and S–P travel times with events relocated in 3D, and validated on held-out events and an independent tomography. The calibration shows that the table rock is too slow near the surface: V_0 is raised by 31%. It also shows that the regional Vs beneath Rainier is 5–9% too slow at 2–7 km, a correction the iMUSH tomography confirms locally.
- **Surface.** Layers for geology, ice, soil, water, vegetation and imagery on a common grid.
- **Geodesy and load.** GNSS strain rates, refreshed weekly, and the stress of the edifice load at depth.

Every input is fetched from its original archive and checksummed. Every parameter names its source, and every product can be downloaded with one command in the formats that seismological codes read. The mass-movement catalogue is a first time-dependent layer; the next are the events digitised from the literature and couplings between the hydrological layers and the seismic properties.

12 Code and data availability

The code, configuration, source registry and this paper are at <https://github.com/Denolle-Lab/mt-rainier-digital-model> (BSD-3-Clause). The derived products are the assets of the release `products-v1.0.0` of the same repository (CC-BY 4.0): the model, the uniform grids, the strain in the volume, the edifice-load stress, the alteration field, the mass-movement catalogue and a GNSS snapshot. They are listed with SHA-256 checksums in `SHA256SUMS` and `src/rainier3d/products.json`, downloaded with `rainier3d fetch`, and described product by product, with command-line and Python examples, in `docs/products.md`. The software and the derived products are deposited on Zenodo as two versioned records (`docs/doi.md`).

Appendix A lists the input data sets with their DOIs or service addresses. `docs/data_policy.md` gives the licence tier of each. `docs/data_manifest.csv` gives the checksum of each cached input.

Two inputs require attribution notices:

- GPS time series are provided by the Pacific Northwest Geodetic Array, Central Washington University.
- The imagery layers contain modified Copernicus Sentinel data (2023, 2025).

The code and this paper were written with an AI coding assistant (Claude, Anthropic) under the direction of the authors. All numbers are produced by the scripts named in the text, and all DOIs were resolved against the Crossref and DataCite registries.

Appendix A: Data sets

Table A1: Input data sets.

Data set	Role in the model	Reference
Cascadia velocity model v1.7	regional Vp and Vs to 9.9 km	Wirth et al. (2025); Wirth Moriarty et al. (2025)
CRESCENT Gen0 community velocity model	regional Vs below 9.9 km	He et al. (2026); He (2026)
Washington 1:100,000 surface geology (GeMS)	surface units	Washington Geological Survey (2026a)
Geology of Mount Rainier National Park	stratigraphy; map overlay	Fiske et al. (1963)
USGS 3D Elevation Program	elevation	U.S. Geological Survey (2026c)
IceBoost v2 per-glacier thickness; RGI 6.0	ice thickness; glacier identifiers	Maffezzoli (2025); RGI Consortium (2017)
GlaThiDa; 1981 radar surveys	glacier thickness check	World Glacier Monitoring Service (WGMS) (2020); Driedger and Kennard (1986)
1996 helicopter EM and magnetic survey	alteration	Rystrom et al. (2000)
iMUSH local-earthquake tomography	independent check	Incorporated Research Institutions for Seismology (2020); Ulberg et al. (2020)
PNSN origins and phase picks (ComCat)	calibration and validation	U.S. Geological Survey (2026a)
FDSN station metadata	station coordinates, sensor inventory	EarthScope Consortium (2026)
PANGA and UNR GNSS daily positions	velocities, strain	Pacific Northwest Geodetic Array (2026); Blewitt et al. (2018)
SOLUS100	soil thickness	Nauman (2024)
US water-table depth	water table	Ma et al. (2026a)
Global water-table depth	water table, comparison	Fan et al. (2017)
NHDPlus High Resolution	streams	U.S. Geological Survey (2026b)
ETH global canopy height 2020	canopy height	Lang, Nico et al. (2022)
NLCD 2021	land cover	U.S. Geological Survey (2023)
Copernicus Sentinel-2 L2A	imagery, NDVI, NDSI, leaf area index	European Space Agency (2025); European Space Agency (2023)

Data set	Role in the model	Reference
GEDl L2B, L3, L4B	plant area index, canopy height, biomass	Dubayah et al. (2021b); Dubayah et al. (2021a); Dubayah et al. (2025)
Washington State Landslide Inventory Database	landslide and debris-flow catalogue	Washington Geological Survey (2026c)
Seismogenic mass movements, western United States	dated, seismically located events	Allstadt et al. (2017)
Mount Rainier volcano hazards, digital data	lahar inundation zones	Hoblitt et al. (1998); Schilling et al. (2008)
Quaternary active faults of Washington	faults	Washington Geological Survey (2026b)

Author contributions. MD designed the model, the calibration and the validation and directed the work. DY designed and built the three-dimensional viewer. MKö produced the vegetation products of the canopy-storage project (lidar canopy height and cover, Sentinel-2 leaf area index and the gridded GEDI products). MH, SH and MKi contributed to the project.

Competing interests. The authors declare that they have no conflict of interest.

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